

Secure Communications

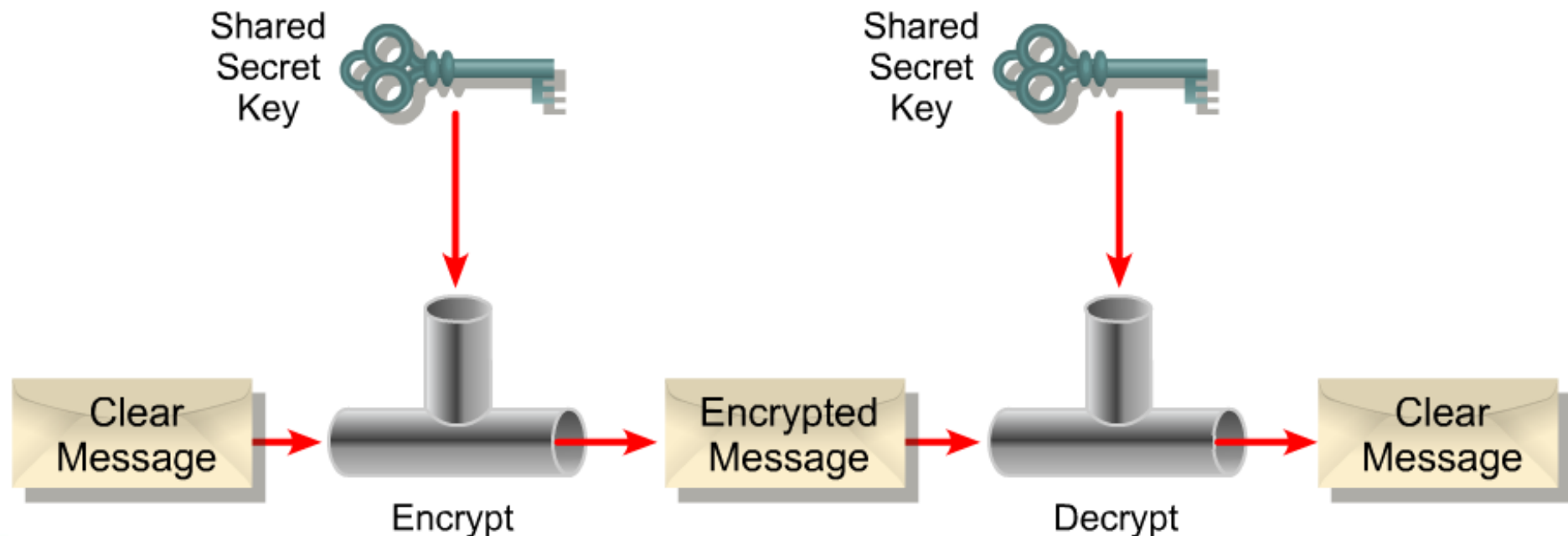
Segurança em Redes de Comunicações
Mestrado em Cibersegurança
Mestrado em Engenharia de Computadores e
Telemática
DETI-UA

Introduction to Cryptography



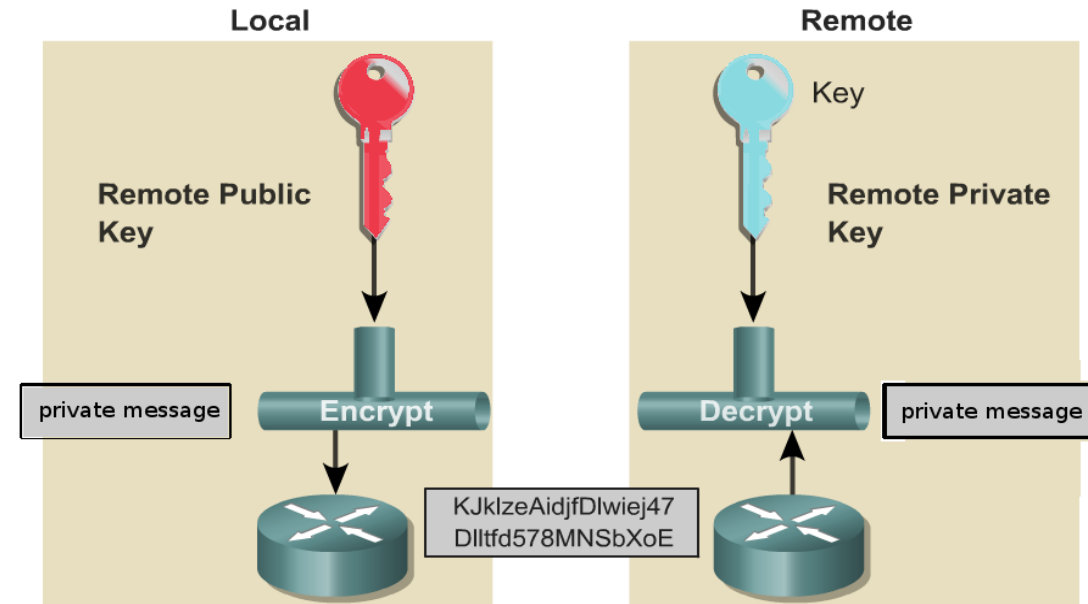
Symmetric Key Cryptography

- Two requirements for secure use of symmetric encryption:
 - Strong encryption algorithm
 - Secret key known only to sender / receiver
- Assume encryption algorithm is known
- Implies a secure channel to distribute key



Public Key Cryptography

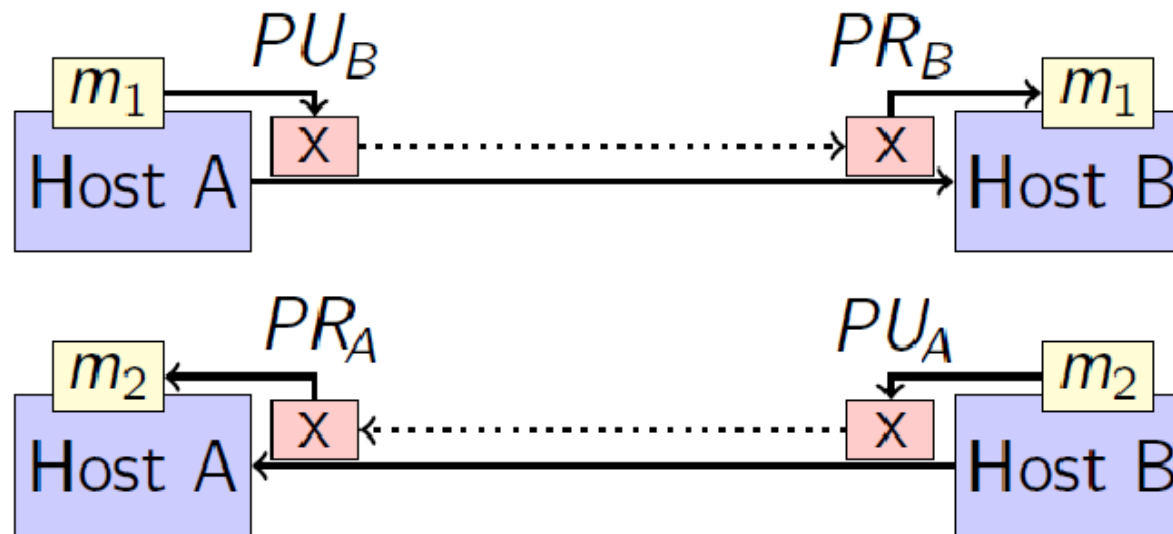
- Public Key Cryptography involves a pair of keys
- **A public key**
 - May be known by anybody, and can be used to encrypt messages, and verify signatures
- **A private key**
 - Known only to the recipient, used to decrypt messages, and sign (create) signatures
- Each public key is published, and the corresponding private key is kept secret
- Is asymmetric because those who encrypt messages or verify signatures cannot decrypt messages or create signatures



Public Key

Encryption for confidentiality

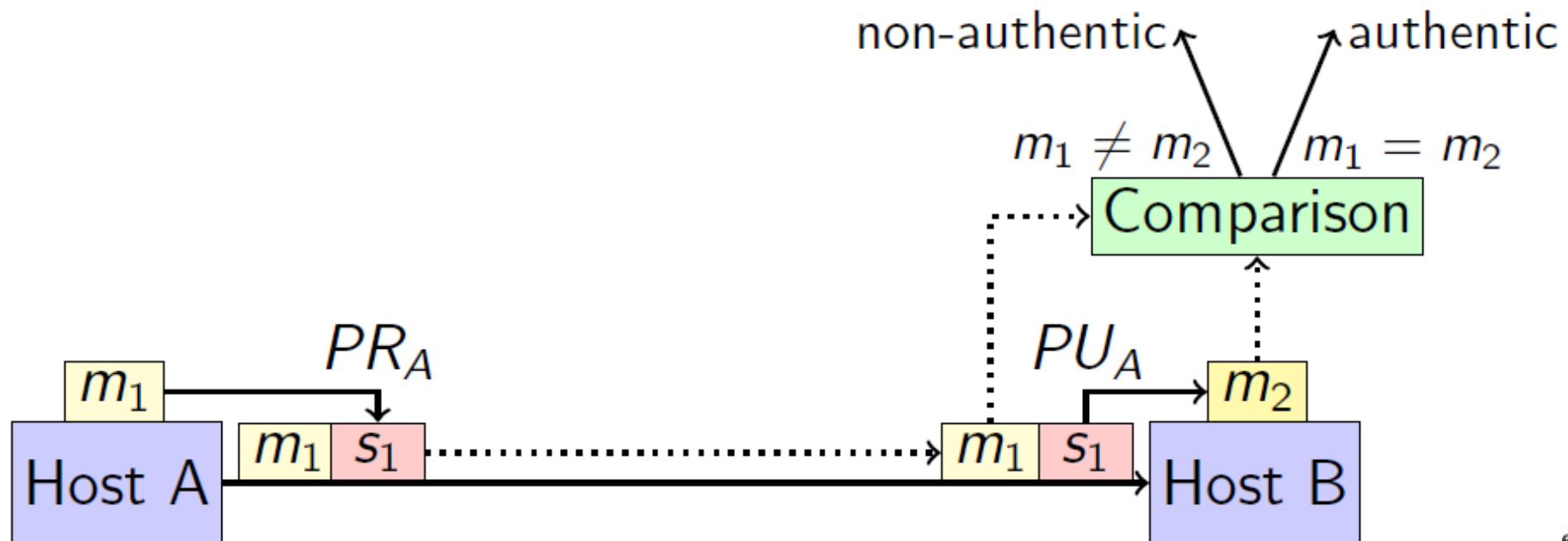
- To send an encrypted message from A to B
 - ♦ Host A encrypts data with Host B public key (PUB)
 - ♦ Host B decrypts data with Host B private key (PRB)
- To send an encrypted message from B to A
 - ♦ Host B encrypts data with Host A public key (PUA)
 - ♦ Host A decrypts data with Host A private key (PRA)
- This method is computational inefficient for encrypting large amounts of data.
- Commonly used to create secure communication channels where a temporary symmetric key can be negotiated and used to encrypt large amounts of data.



Public Key

Digital signatures for authentication

- To send an authenticated message from A to B
 - Host A creates a signature by encrypting data with Host A private key (PR_A)
 - Host A sends data and signature to host B
 - Host B verifies data by decrypting signature with Host A public key (PU_A) and compares with received message



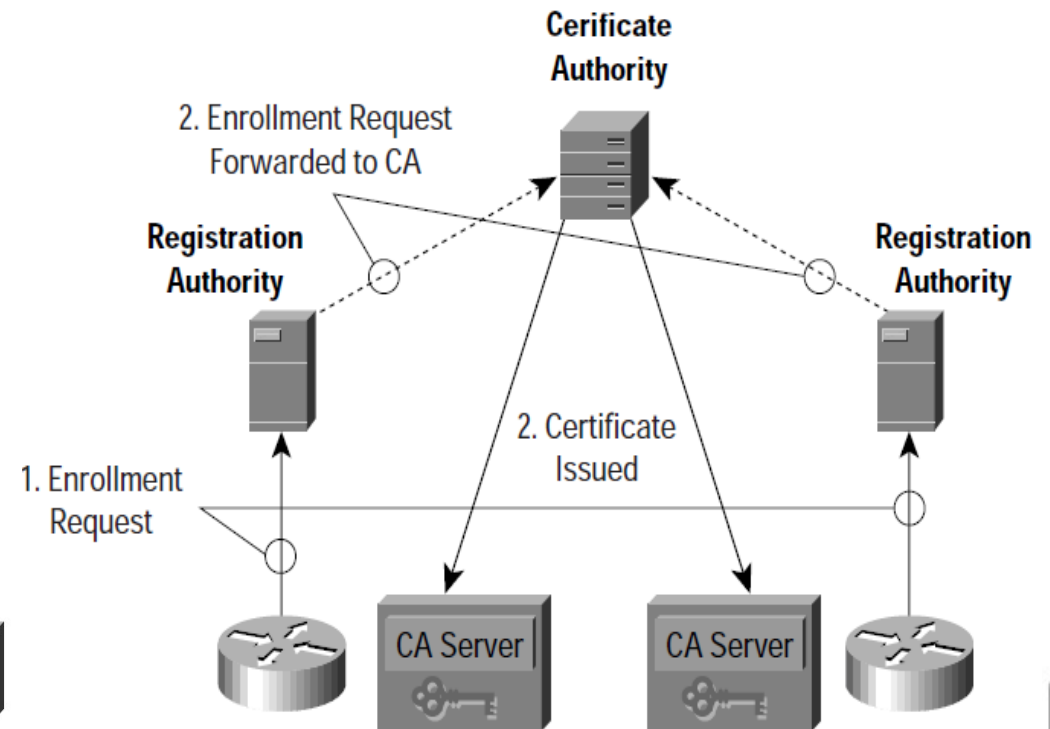
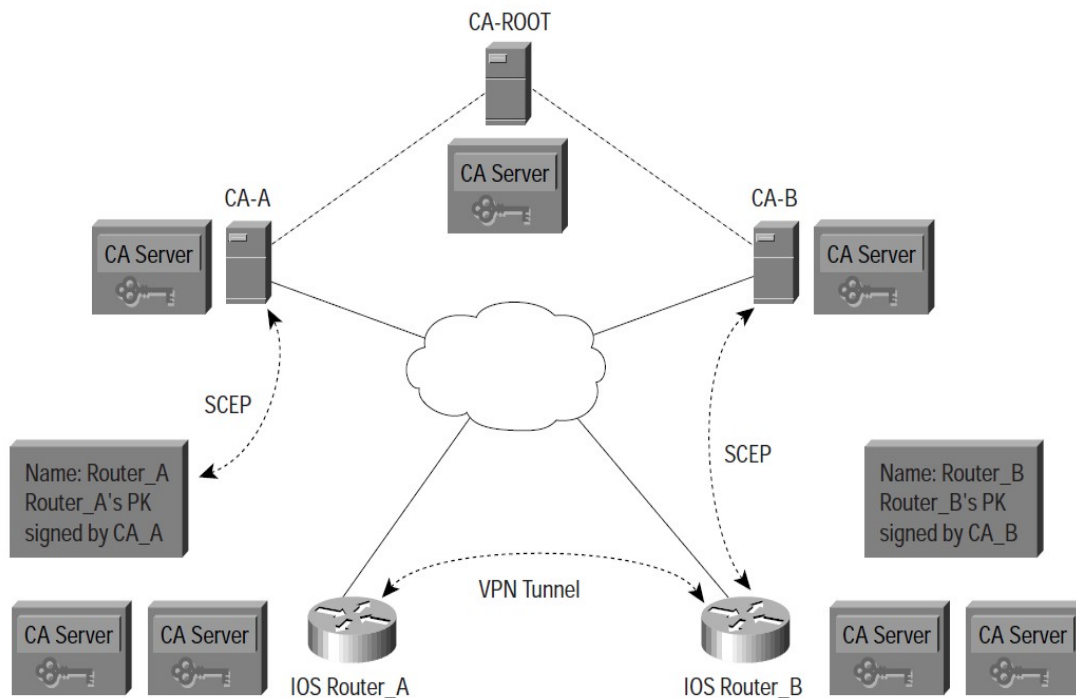
RSA (Rivest, Shamir and Adleman)

- Named after its inventors - Rivest, Shamir and Adleman
- It's a public key algorithm (encryption and decryption)
- Key length is variable
 - Common key lengths: 512, 1024 and 2048 bits
- Block size is variable but must be smaller than key length
- Ciphertext length will be the length of the key
- Slower than DES, AES and IDEA
 - Usually not used to encrypt large messages
 - Used to encrypt secret key and secret key used to encrypt messages



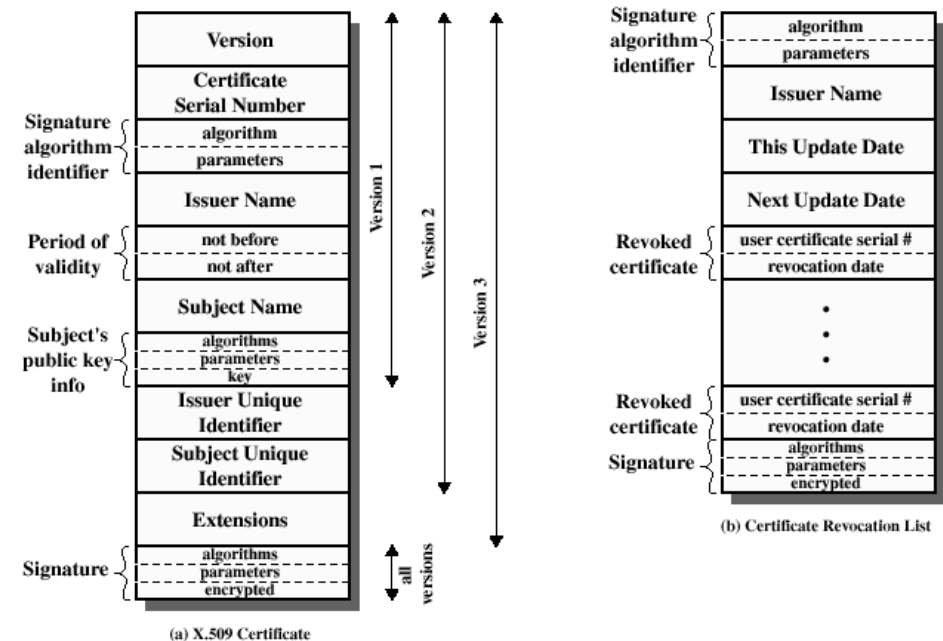
Public Key Infrastructure (PKI)

- PKIs are hierarchical in nature
- Each PKI participant holds a digital certificate that has been issued by a Certificate Authority (CA)
 - CA may be a root CA or a subordinate CA
 - Trust chain.
- PKI might use additional hosts called Registration Authorities (RA) to accept requests for enrollment in the PKI



X.509 certificate contents

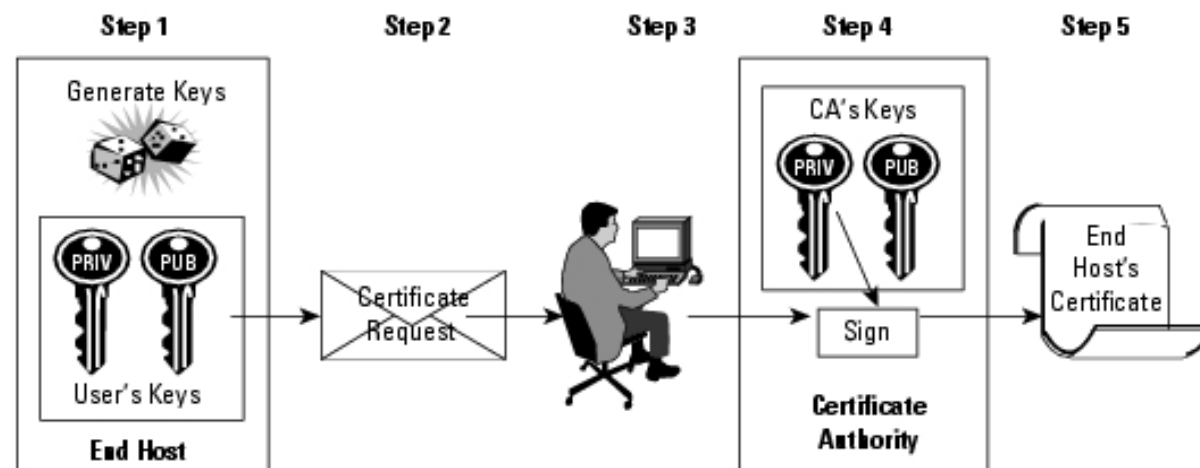
- Version
- Serial Number
- Signature Algorithm
- Issuer Name
- Validity Period
- Subject Name



- Subject Name
 - Distinguished Name (DN) of the entity
 - CN=Java Duke, OU=Java Software Division, O=U.Aveiro, C=PT
- Subject Public Key Information
 - Public Key Algorithm
 - Subject Public Key
- Certificate Signature Algorithm
- Certificate Signature

Certificate Authority enrollment

- Simple Certificate Enrollment Protocol (SCEP) is used for secure transportation of key information and certificates
- Enrolling in a Certificate Authority
 1. End host generates a private-public key pair
 2. End host generates a certificate request, which it forwards to the CA
 3. Manual, human intervention is required to approve the enrollment request,
 4. After the approval, the CA signs the certificate with its private key and returns the completed certificate to the end host
 5. End host stores certificate

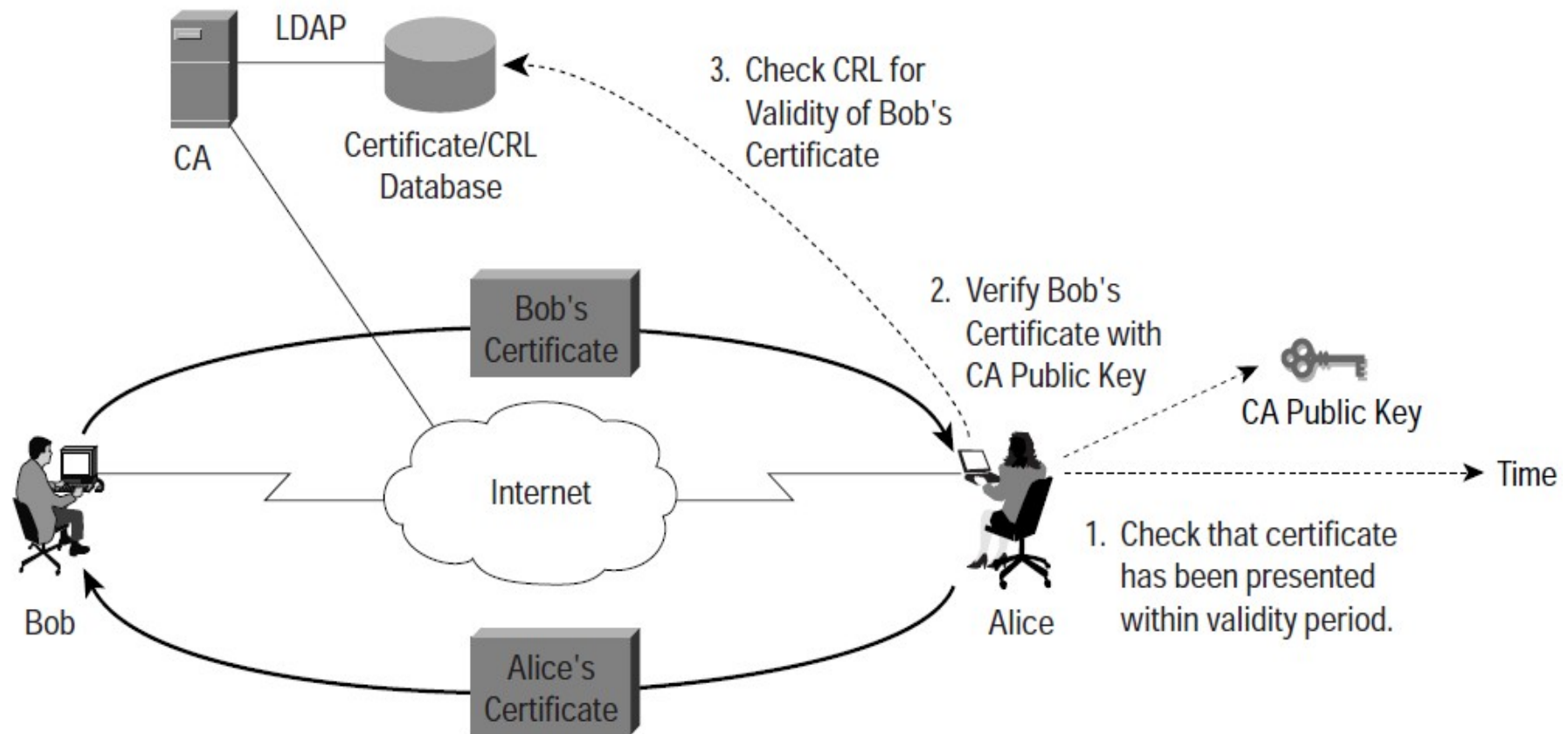


Certificate Revocation Lists (CRL)

- The CRL is another crucial PKI component
- Is a list of certificates that were formerly valid within the PKI, but have been revoked for some reason
- These reasons could include any of the following:
 - Compromise of keys within certificate
 - Loss of access privileges for user/device
 - Change of PKI structure requiring certificate re-issue



Certificate usage and validity check



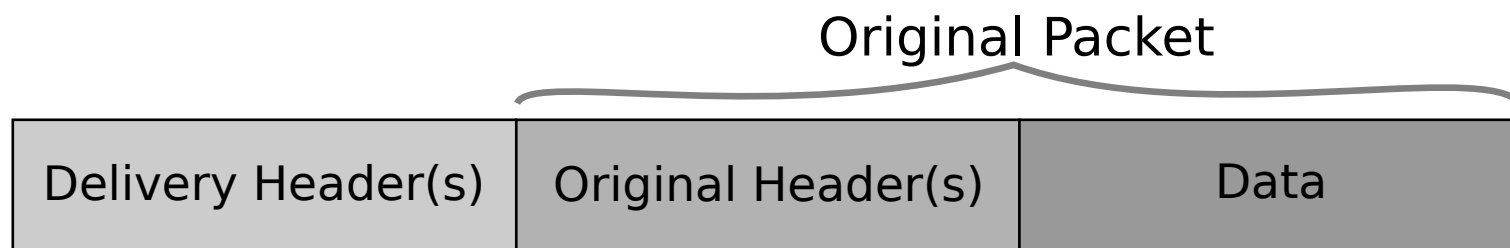
- The cert is being presented within its validity period
- The CA that signed the cert is known and trustable
- The certificate is not on a revocation list (optional in some scenarios)

Virtual Connections

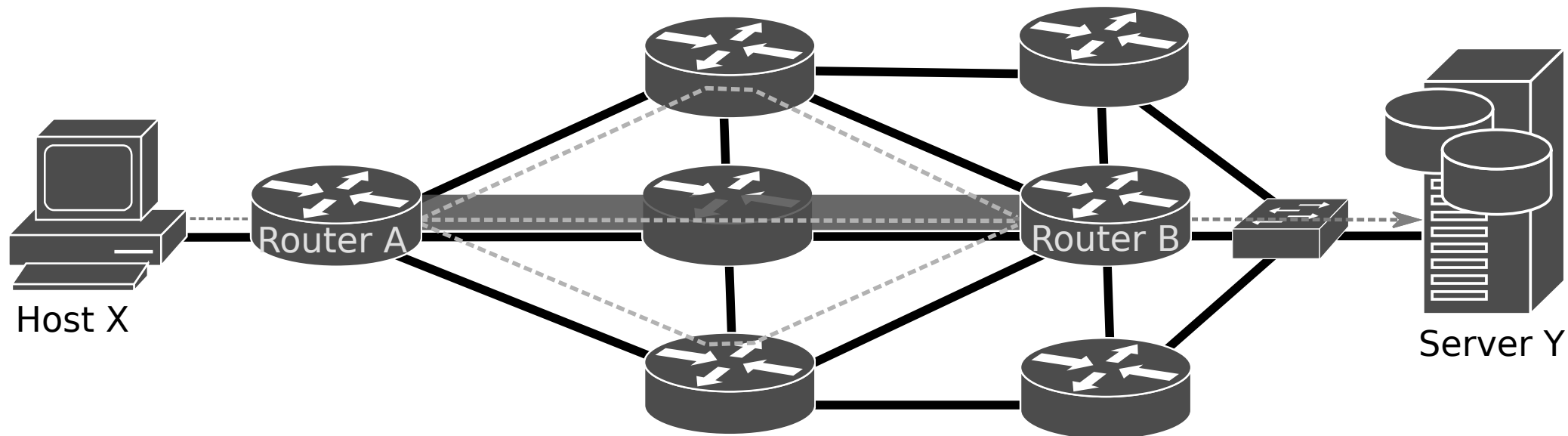


Traffic Tunnel Concept

- Main purposes
 - Guarantee that a packet that reaches a network node will reach a specific secondary network node independently of the intermediary nodes routing processes,
 - Guarantee the delivery of a packet to a remote node when the intermediary nodes do not support the original packet network protocol, and,
 - Define a virtual channel that adds additional data transport features in order to provide differentiated QoS, security requirements and/or optimized routing.
- Achieved by adding, at the tunnel entry point, one or more protocol headers to the original packets to handle their delivery to the tunnel exit point.



Tunnel End-Points



Delivery protocol(s)	Original protocol(s)	Data
Source: A address Destination: B address	Source: X address Destination: Y address	

Virtual Tunnel Interface (VTI)

- Logical construction that creates a virtual network interface that can be handled as any other network interface within a network equipment.
- A tunnel does not require to have any network addresses other the ones already bound to the end-point router.
- However, most implementations impose that a network address must be bound to a tunnel interface in order to enable IP processing on the interface.
 - ◆ The tunnel interface may have a explicitly bound network address or reuse an address of another interface already configured on the router.

```
1 #interface Tunnel 1
2 #ip address 10.1.1.1 255.255.255.252
3 #ipv6 address 2001:A:A::1/64
4 #ip unnumbered FastEthernet0/0
5 #ipv6 unnumbered FastEthernet0/0
6 #ip ospf cost 10
7 #ipv6 ospf 1 area 0
8 #tunnel mode ipip
9 #tunnel source FastEthernet0/0
10 #tunnel destination 200.2.2.2
```



VTI Requirements

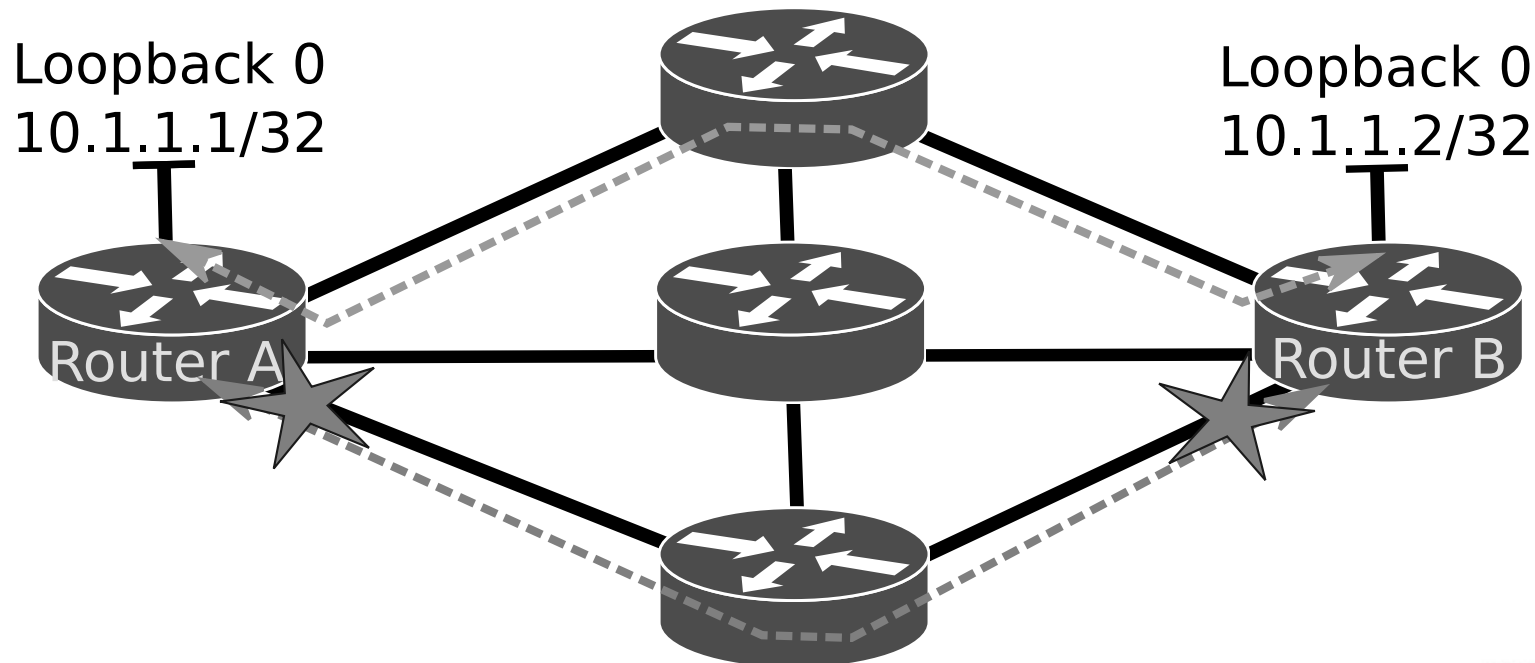
- A numeric identifier,
- A bounded IP address, this will enable IP processing,
 - Add the tunnel interface to the routing table and allow routing via the interface,
- A defined mode or type of tunnel,
 - Availability of tunnel models depends on the Router model, operating software and licenses.
- Tunnel source,
 - Defined as the name of the local interface or IPv4/IPv6 address depending on the type of the tunnel.
- Tunnel destination,
 - Defined as a domain name or IPv4/IPv6 address depending on the type of the tunnel.
 - This definition is not mandatory for all types of tunnels because in some cases the tunnel end-point is determined dynamically.
- May optionally have additional configurations for routing, security and QoS purposes.

```
1 #interface Tunnel 1
2 #ip address 10.1.1.1 255.255.255.252
3 #ipv6 address 2001:A:A::1/64
4 #ip unnumbered FastEthernet0/0
5 #ipv6 unnumbered FastEthernet0/0
6 #ip ospf cost 10
7 #ipv6 ospf 1 area 0
8 #tunnel mode ipip
9 #tunnel source FastEthernet0/0
10 #tunnel destination 200.2.2.2
```



Loopback Interfaces as End-Points

- Loopback interface is another logical construction that creates a virtual network interface completely independent from the remaining physical and logical router network interfaces.
- The main propose of a loopback interface is to provide a network address to serve as router identifier in remote network configurations and distribute algorithms.
- The main advantage of using loopback interfaces as tunnel end-points, is the creation of a tunnel not bounded to any individual network card/link that may fail.



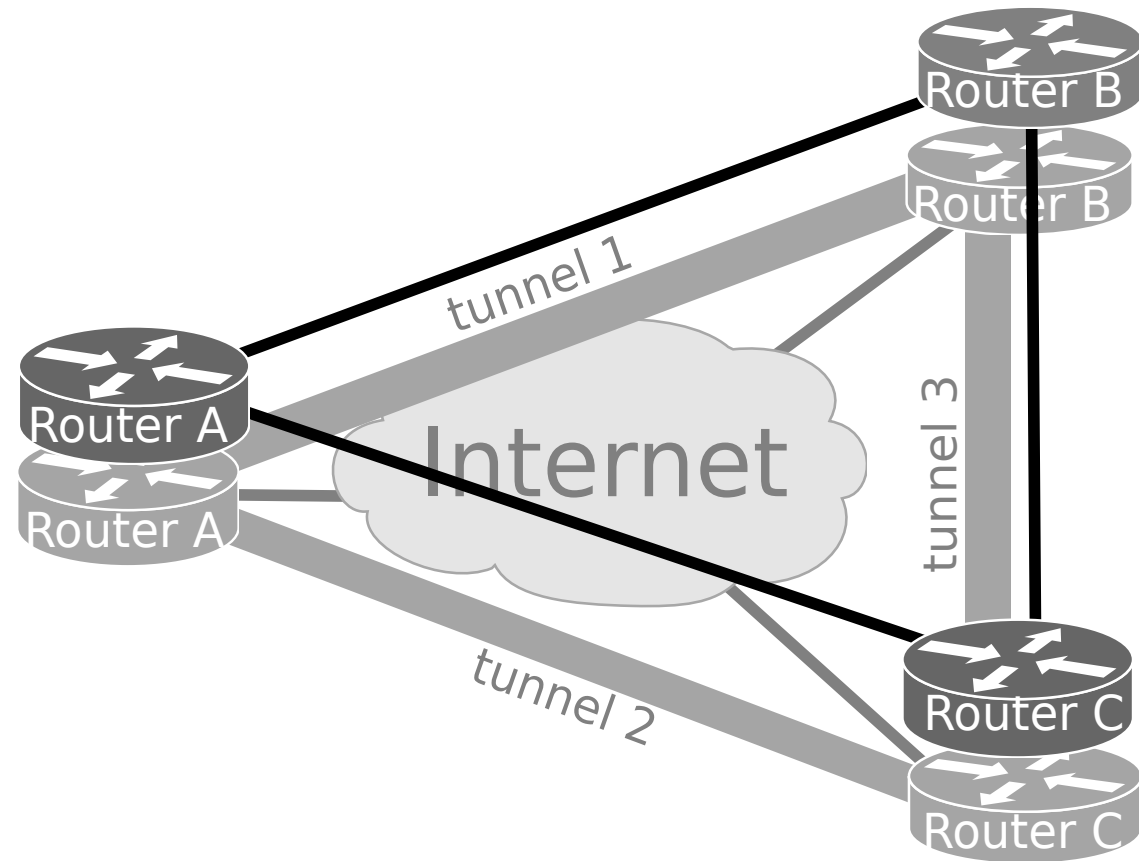
IP Tunnel Types

- IPv4-IPv4
 - ♦ Original IPv4 packets are delivered using IPV4 as network protocol.
- GRE IPv4
 - ♦ Original packets protocol (any network protocol) is defined by GRE header and delivered using IPv4 as network protocol.
- IPv6-IPv6
 - ♦ Original IPv6 packets are delivered using IPv6 as network protocol.
- GRE IPv6
 - ♦ Original packets protocol (any network protocol) is defined by a GRE header and delivered using IPv6 as network protocol.
- IPv6-IPv4
 - ♦ Original IPv6 packets are delivered using IPv4 as network protocol.
- IPv4-IPv6
 - ♦ Original IPv4 packets are delivered using IPv6 as network protocol.

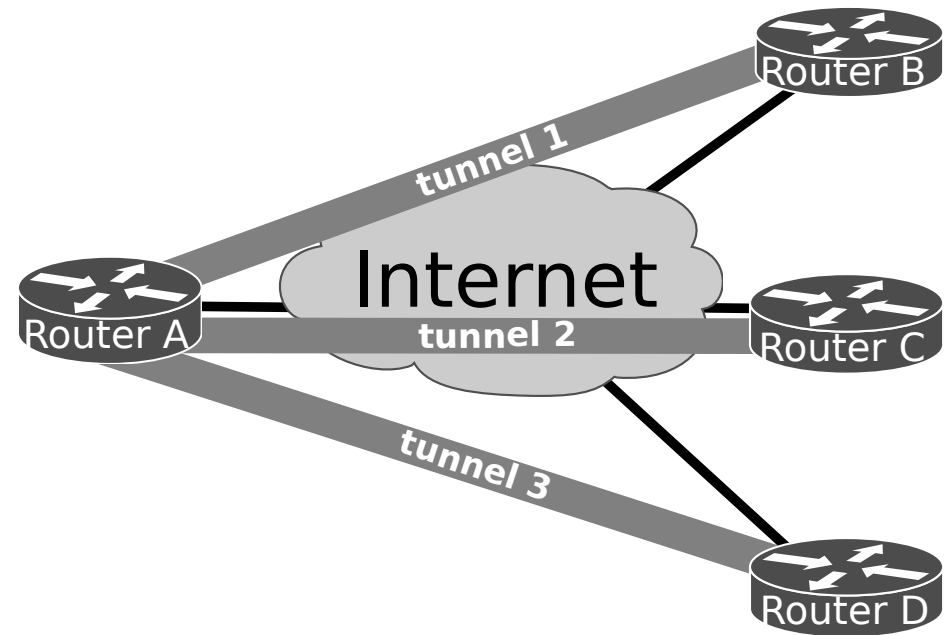
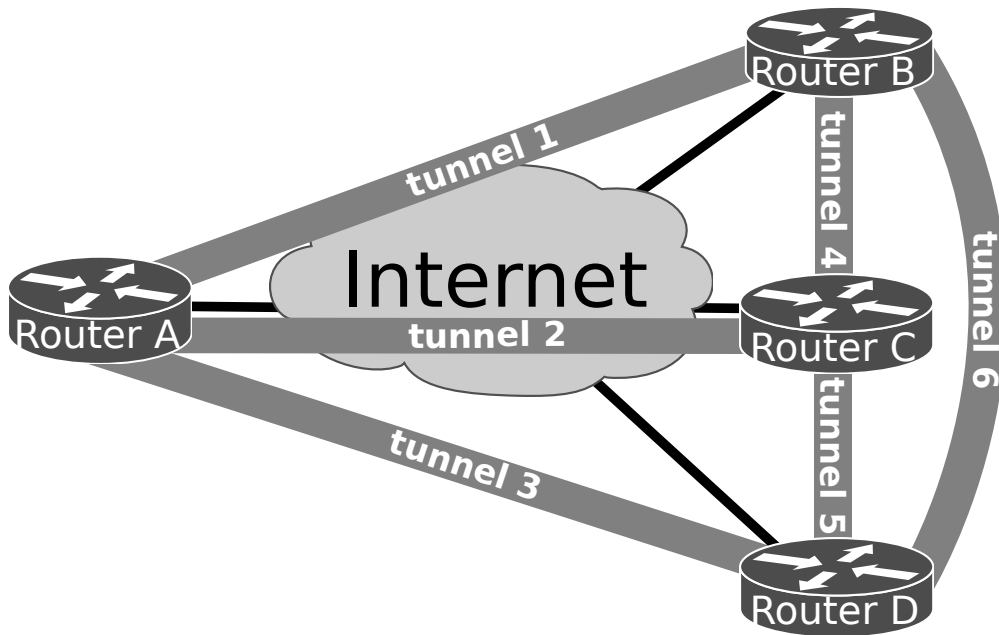


Overlay Network

- An overlay network can be defined as a virtual network defined over another network.
 - For a specific purpose like private transport/routing policies, QoS, security.
- The underlying network can be physical or also virtual.
 - May result in multiple layers of overlay networks.
- When any level of privacy protocol is present on an overlay network is designated by Virtual Private Network (VPN).



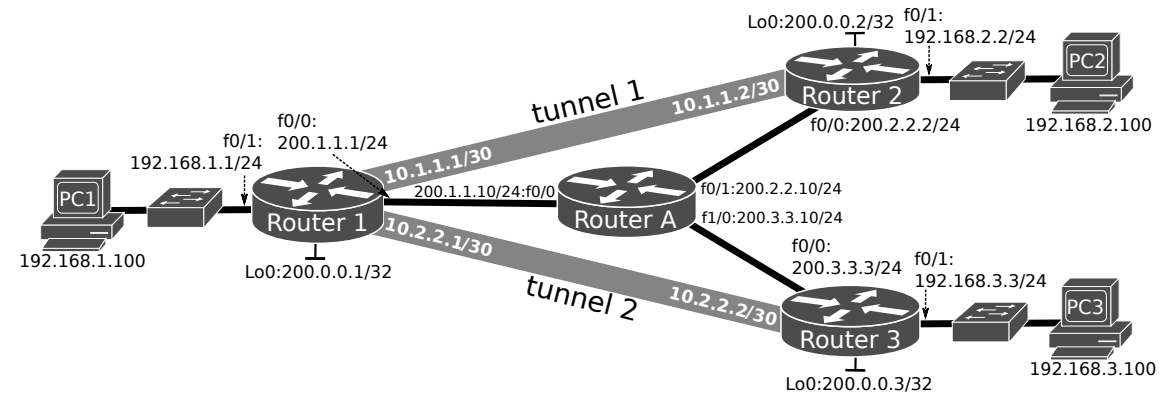
Full/Partial Overlay Mesh



Routing Through/Between Tunnels

- Static Routes

```
1 #ip route 192.168.2.0 255.255.255.0 Tunnel1
2 #ip route 192.168.2.0 255.255.255.0 10.1.1.2
3 #ipv6 route 2001:A:1::/64 Tunnel1
4 #ipv6 route 2001:A:1::/64 2001:0:0::2
5 #ip route 192.168.2.100 255.255.255.255 10.1.1.2
6 #ipv6 route 2001:A:1::100/128 2001:0:0::2
```



- Policy Based Routing (route-maps)

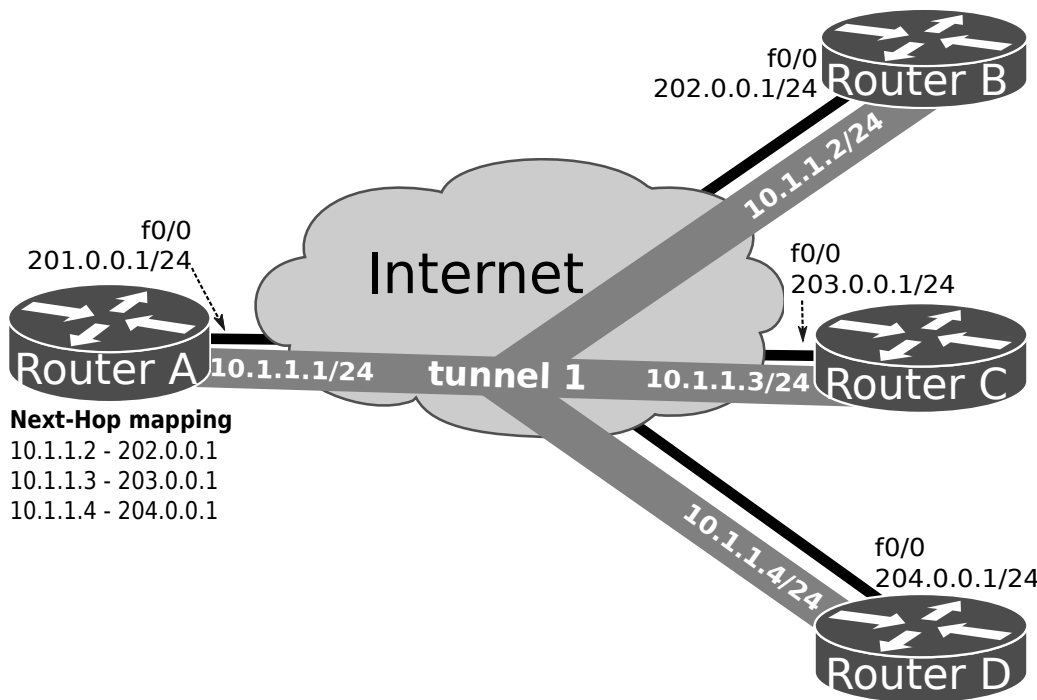
```
1 #access-list 100 permit ip host 192.168.1.100 192.168.2.0 255.255.255.0
2 #route-map routeT1
3   #match ip address 100
4   #set ip next-hop 10.1.1.2
5 #interface FastEthernet0/1
6   #ip policy route-map routeT1
```

- Dynamic Routing

- Multiple (distinct) routing processes.
 - One per overlay network, and
 - One for the underlying network.

```
1 #router ospf 1
2   #network 200.1.1.0 0.0.0.255 area 0
3   #network 200.0.0.1 0.0.0.0 area 0
4   !
5 #router ospf 2
6   #network 10.0.0.0 0.255.255.255 area 0
7   #network 192.168.0.0 0.0.255.255 area 1
```

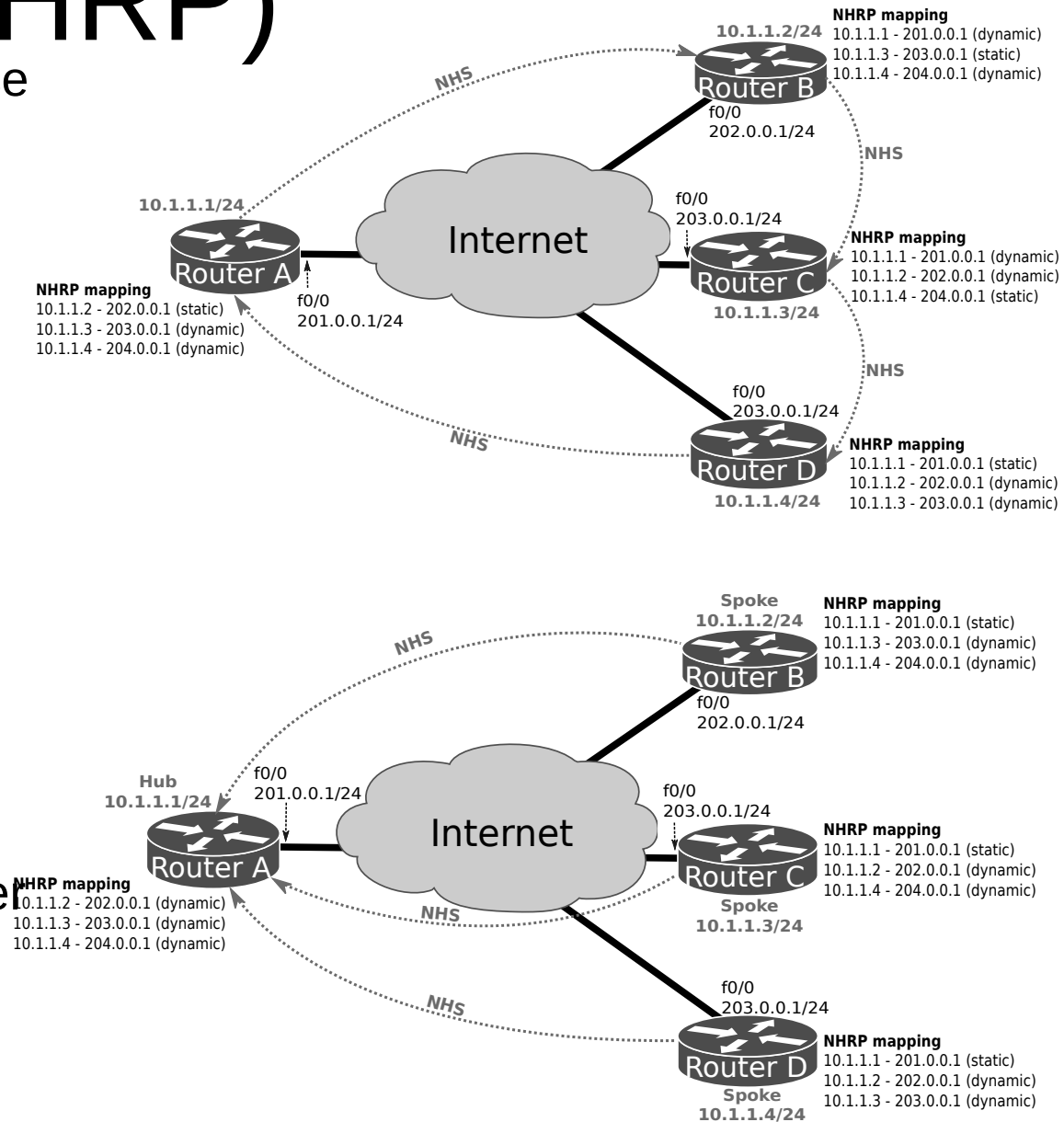

Multipoint Tunnels



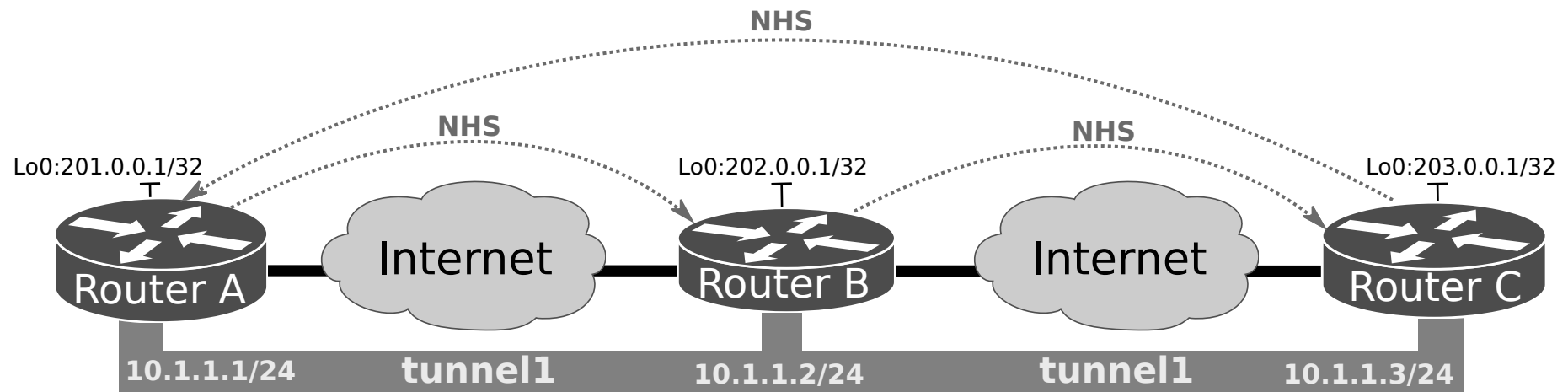
- In a scenario with many nodes to interconnect, the simpler and more efficient approach is to have a single tunnel that interconnect multiple nodes - a multipoint tunnel.
- Directly connect using a single virtual overlay IP network, defined within a multipoint tunnel.
- In a multipoint tunnel scenario, the delivery header address is determined based on the address of the next hop within the overlay network.
- Address mapping between overlay and underlying network addresses may be statically defined or dynamically obtained.

Next Hop Resolution Protocol (NHRP)

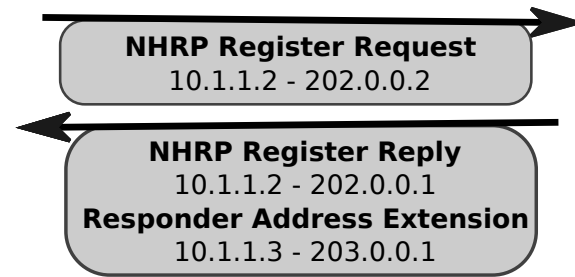
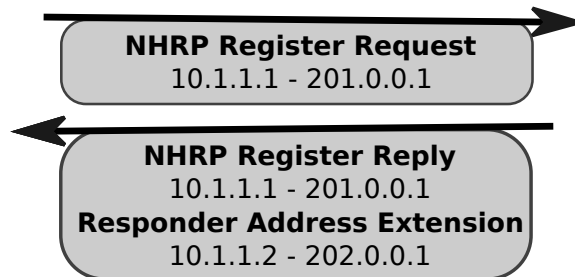
- NHRP allows to map a tunnel interface IP address (overlay network) to the respective underlying network interface IP address.
- NHRP tunnel requires that all intervening nodes should be able to find a path to any of the other nodes.
- Each node should at least know one other overlay node (and respective overlay and underlaying addresses) through which he will try to find the other nodes address mappings.
 - Next Hop Server (NHS).
- Moreover, all nodes must be configured in a way that all nodes have at least one valid path to all other nodes - forming a partial mesh.



NHRP Information Exchange

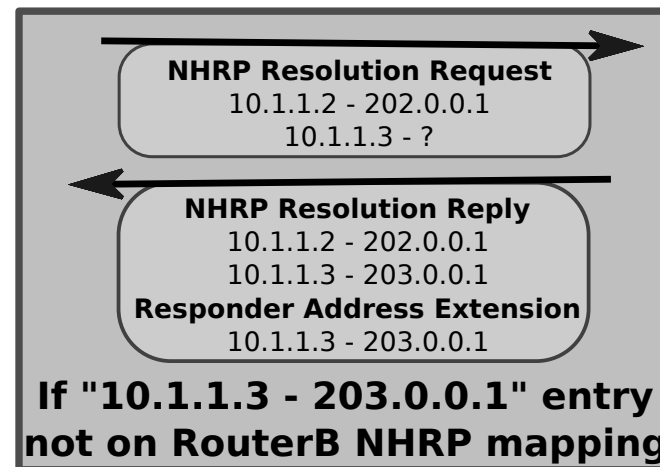
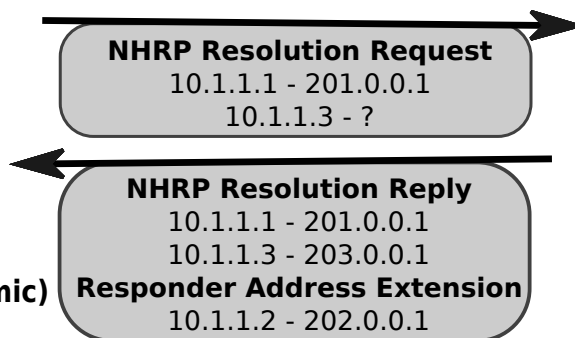


NHRP mapping
10.1.1.2 - 202.0.0.1 (static)



NHRP mapping
10.1.1.1 - 201.0.0.1 (static)

NHRP mapping
10.1.1.1 - 201.0.0.1 (static)
10.1.1.2 - 202.0.0.1 (dynamic)



NHRP mapping
10.1.1.2 - 202.0.0.1 (static)
10.1.1.3 - 203.0.0.1 (dynamic)

Hub-Spoke vs. Spoke-Spoke

- Hub-Spoke

- Each remote site is connected with a point-to-point GRE tunnel to a pre-defined central node (Hub).
- Hub accepts new tunnel connections from Spokes (branches nodes).
- Data communication (over the overlay network) between Spokes is relayed via the Hub.
- Multiple Hubs may exist to provide redundancy.

- Spoke-Spoke

- Individual branch office nodes can dynamically initiate tunnel connections between each other, bypassing the Hub node.
- Data communication (over the overlay network) can be direct between Spokes.
- Dynamic IGP routing protocols may operate between Spoke and Hubs, but not between Spokes.
- No interoperability with non-Cisco IOS routers. (?)

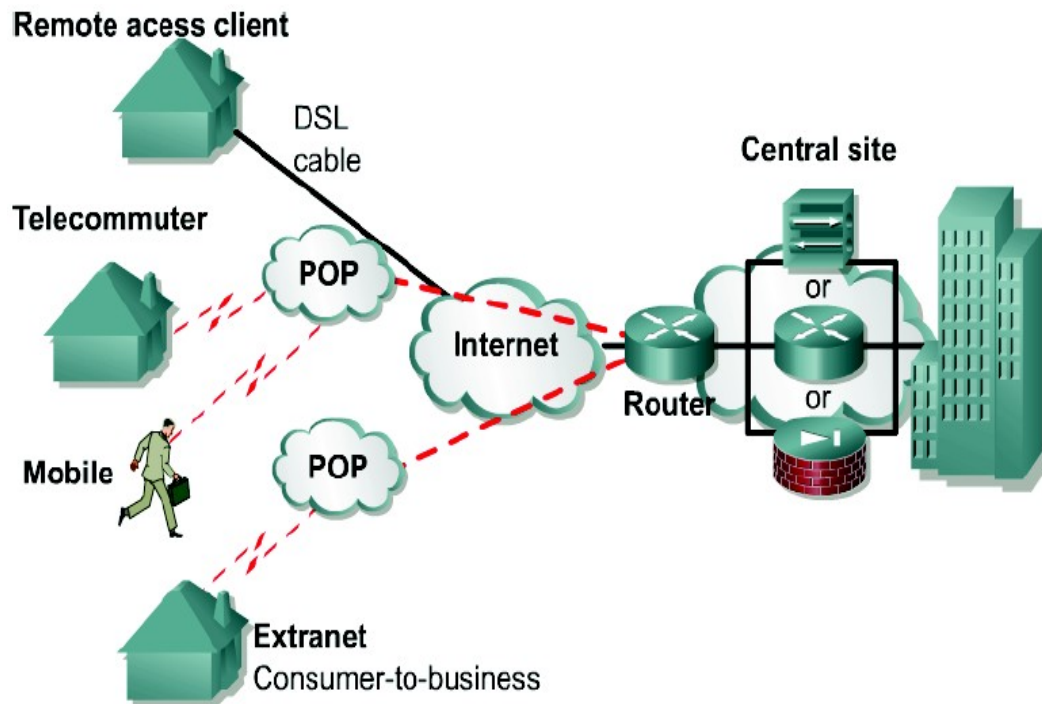


Virtual Private Networks (VPN)

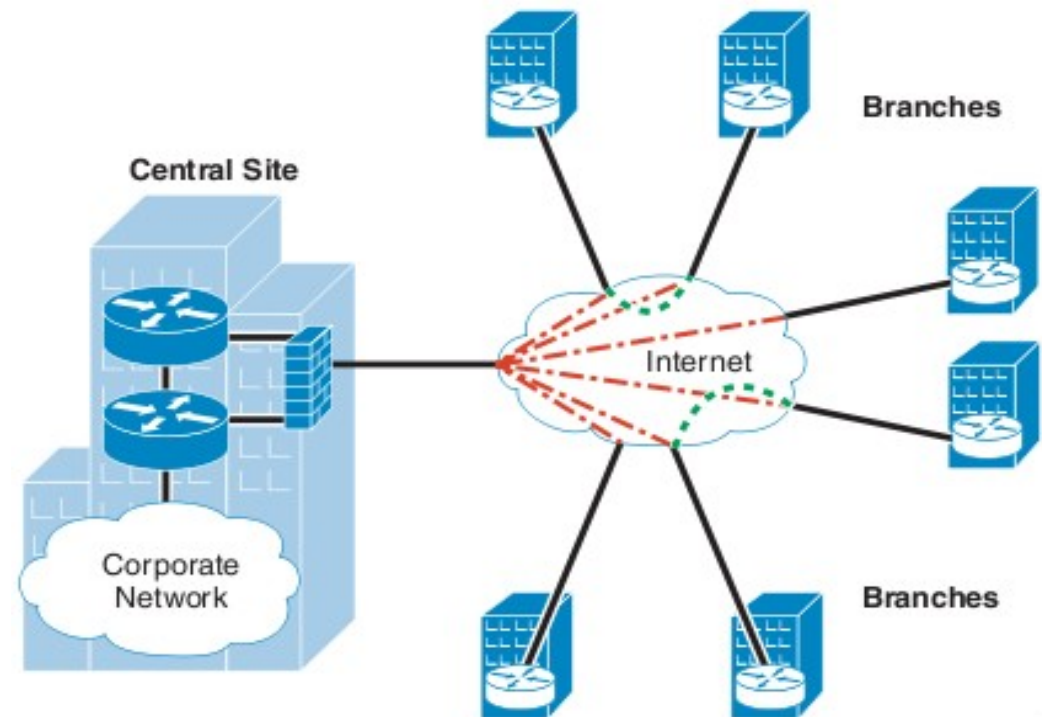


VPN - Virtual Private Networks

- Is an encrypted connection between private networks over a public network



- Remote Access VPN



- Site-to-Site VPN

Remote Access VPNs

Protocols and integration



Remote Access (1)

- Most common servers/protocols
 - L2TP IPsec
 - IKE+ISAKMP+L2TP
 - OpenVPN
 - SSL
 - Wireguard
 - Opensource
 - Tailscale with Wireguard
 - P2P or with relay nodes (DERP).
 - Proprietary
 - SSL or IPSec based.
- Authentication
 - Types
 - Pre-shared
 - RADIUS/LDAP
 - RSA with embedded CA
 - RSA with external CA
 - Certificates/Credentials must be distributed securely
 - Web service, SSH, ...

Remote Access VPN - L2TP/IPSec VPN

- Authentication can be performed with Digital Certificates (RSA) or with the same PPP authentication mechanisms as PPTP
- Provides data integrity, authentication of origin and replay protection
- Encryption provided by IPSec (ESP protocol)
- Can support multiple, simultaneous tunnels for each user
- Slower performance than PPTP.



Remote Access with SSL/TLS

- OpenVPN
 - Implements a SSL/TLS VPN
 - Allows PSK, certificate, and login/password based authentication
 - Encryption provided by OpenSSL (can use all ciphers available)
 - Compatible with dynamic and NAT addresses
- SSL/TLS VPN
 - SSL/TLS protocol handles the VPN tunnel creation
 - SSL/TLS is much easier to implement than IPSec and provides a simple and well-tested platform
 - RSA handshake (or DH) is used exactly as IKE in IPSec
- SSH VPN
 - VPN over a SSH connection
 - SSH tunneling - port forwarding



WireGuard

- Point-to-point secure connection.
- Uses the NoiseIK handshake
 - ◆ Provides mutual authentication and forward secrecy.
- Uses only UDP.
- The server remains silent.
 - ◆ Does not respond to packets unless they are signed
 - ◆ Makes it invisible to port scanners.
- Uses specific high-performance stack
 - ◆ ChaCha20 for symmetric encryption,
 - ◆ Poly1305 for data authentication,
 - ◆ Curve25519 for elliptic curve DH key exchange,
 - ◆ and BLAKE2s for hashing.
- Multi-point connections may be achieved using an Hub-and-Spoke architecture.
 - ◆ With multiple point-to-point connections.
 - ◆ No centralized coordination.



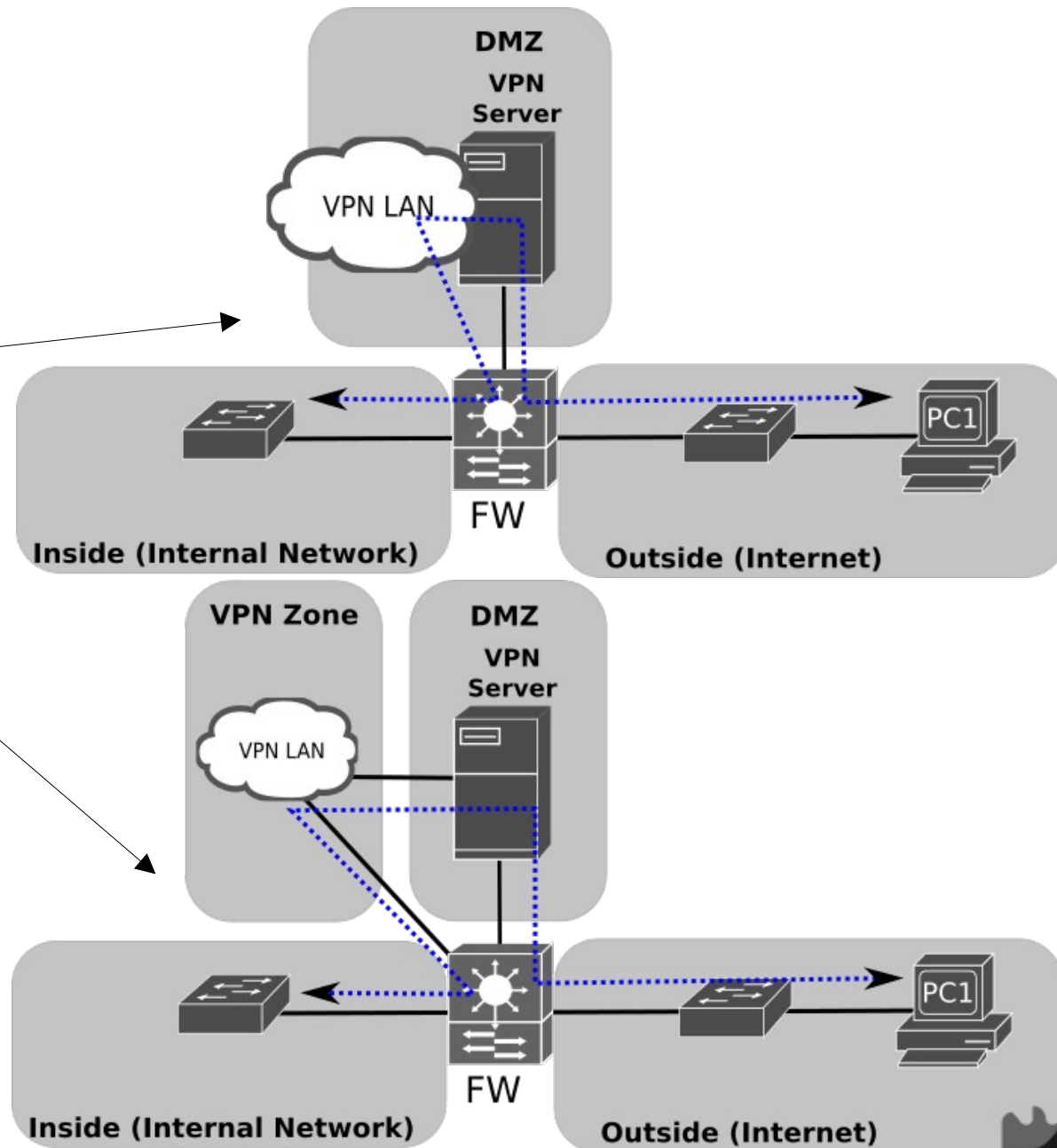
Tailscale

- Multi-point connections are created with a Mesh network with WireGuard.
 - Uses a coordination server to deploy mesh network.
- Unlike WireGuard which identifies end-points by cryptographic keys, Tailscale uses a central coordination server to map those keys to specific identities.
 - SSO (Single Sign-On).
- Automated NAT Traversal (STUN/ICE) and UDP Hole Punching
 - Each private end-point use STUN servers to find it's own public IP address and port;
 - All private end-points exchange public IP/ports via the coordination server;
 - UDP hole punching, relies on the firewall allowing answers to established connections. Sends UDP packet to peers' public IP/port, firewall will allow packets from that IP/port to "answer".
 - Note that all coordination is done via the coordination server.
- When a direct P2P connection is impossible (both end-points behind NAT, and Firewall does not have reflexive rules), it uses relay servers
 - DERP (Designated Encrypted Relay for Packets).



Remote VPN Network Integration

- Server deployed in Firewalls.
- Server in DMZ.
 - Traffic routed back to the firewall using the same zone.
 - Traffic routed back to the firewall using a different network interface and zone.
 - Requires multiple network links/VPNs, and/or
 - Policy based Routing (PBR) in the VPN server.
 - Traffic routed directly to private zone.
 - Breaks zone concept.
 - Unless VPN server implements flow control (a Firewall).



Integration with Flow Control

- Service/protocol rules
 - OpenVPN
 - ➔ Used UDP port.
 - Usually port UDP 1194.
 - IPsec
 - ➔ UDP port 500 for IKE.
 - ➔ IP protocol number 50 (ESP).
 - IP protocol number 51 (AH).
 - ➔ UDP port 4500 for NAT traversal.
 - L2TP
 - ➔ UDP port 1701.
 - ➔ Exception may not be required when L2TP is encapsulated within IPsec packets.
 - Wireguard
 - ➔ UDP port 51820.
 - Tailscale
 - ➔ TCP port 443 and UDP port 3478.
 - ➔ For P2P, UDP port 41641.
- User flows' rules
 - Remote users are assigned a IP network address.
 - Flow control based on IP address or zone.

Site-to-Site VPNs



Site-to-Site VPNs

- IPsec
 - Point-to-point
 - GRE over IPsec, L2TP/IPsec
 - Multipoint (DMVPN)
- SSL/TLS
 - OpenVPN – Point-to-point
 - SoftEther – Point-to-point → LAN to LAN
 - Many proprietary
- Wireguard
 - Point-to-point
 - Multi-point – Hub-Spokes
 - Mesh - Tailscale
- AnySec
 - Provides secure layer to MAC, IP, MPLS, SR, SRv6, others
 - Designed for inter-datacenter connections
- SD-WAN / Cloud WAN
- Non-encrypted (outside of the scope of this course)
 - MPLS VPN
 - EVPN (VXLAN)
 - Can be tunneled over IPsec
 - SR (Segmentation Routing – MPLS labels)
 - SRv6 (Segmentation Routing – IPv6 labels)
 - Can be tunneled over IPsec



IPSec

- Framework of security protocols and algorithms used to secure data at the network layer
- Authentication Header (AH)
 - Ensures data integrity
 - Does not provide confidentiality
 - Provides origin authentication
 - Uses Keyed-hash mechanisms
- Encapsulating Security Payload (ESP)
 - Provides data confidentiality (encryption)
 - Data Integrity
 - Does not protect IP header
- AH and ESP use symmetric secret key algorithms, although public key algorithms are feasible



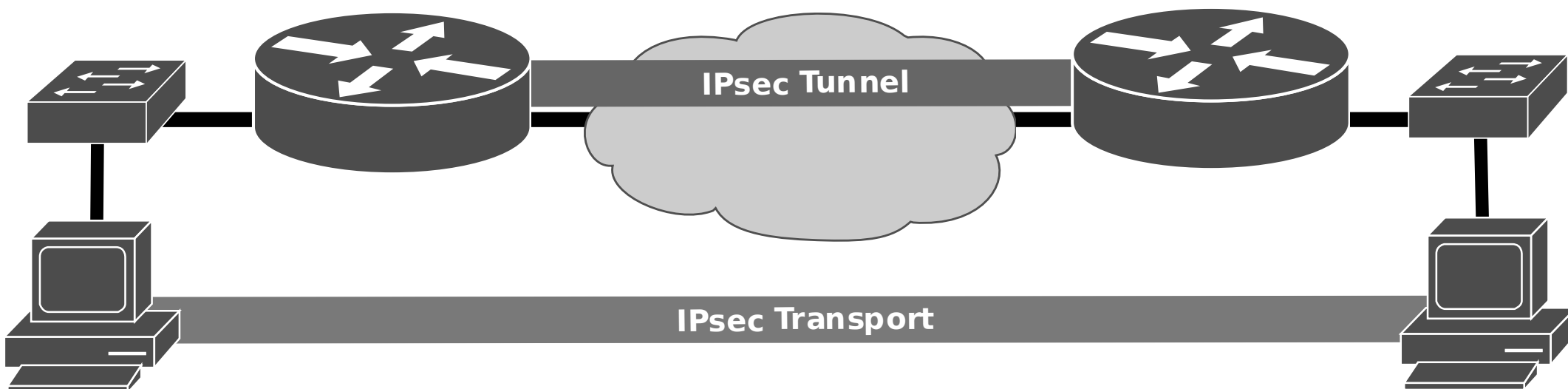
IPSec Modes

- Tunnel

- IPSec gateways provide IPSec services to other hosts in peer-to-peer tunnels
- End-hosts are not aware of IPSec being used to protect their traffic
- IPSec gateways provide transparent protection over untrusted networks

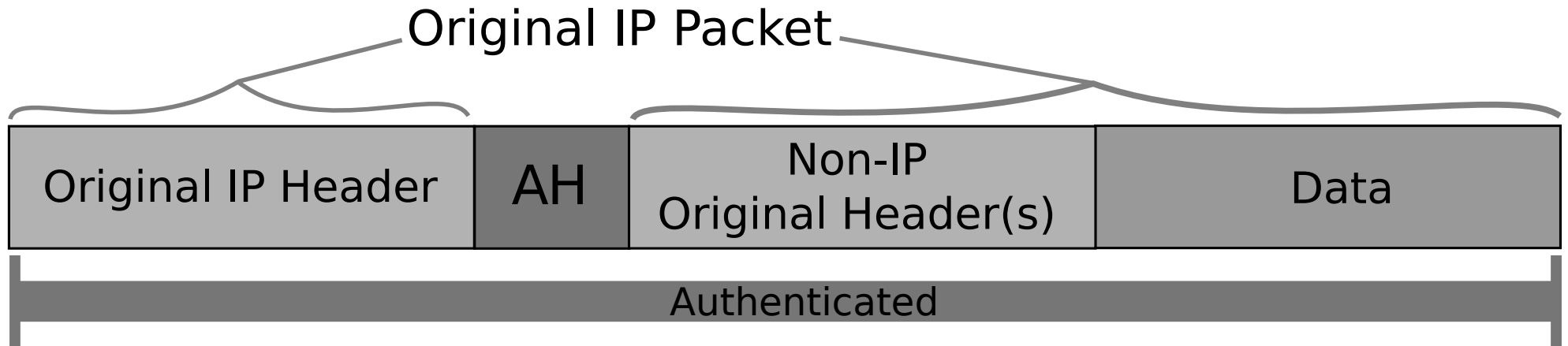
- Transport

- Each end host does IPSec encapsulation of its own data, host-to-host.
- IPSec has to be implemented on end-hosts
- The application endpoint must also be the IPSec endpoint

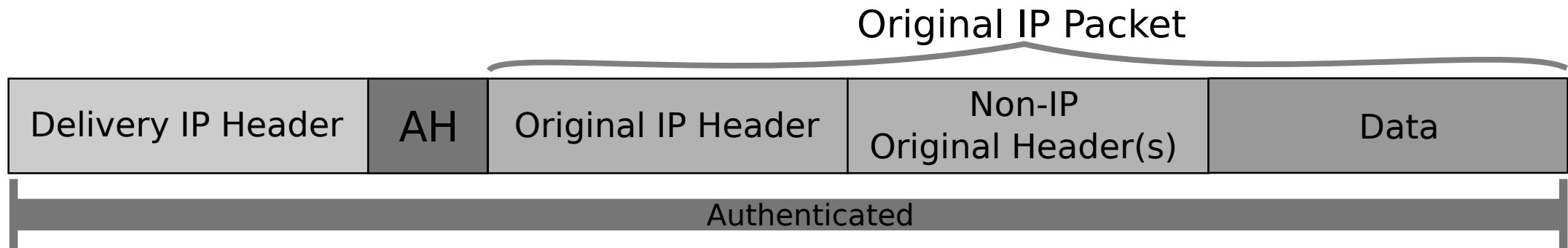


IPSec - AH header placement

- Transport mode

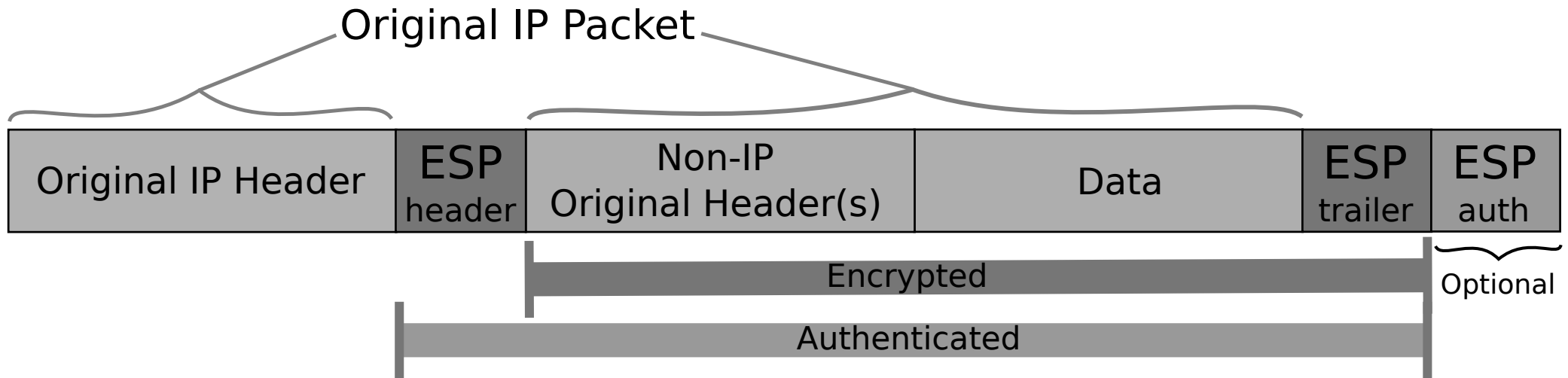


- Tunnel mode

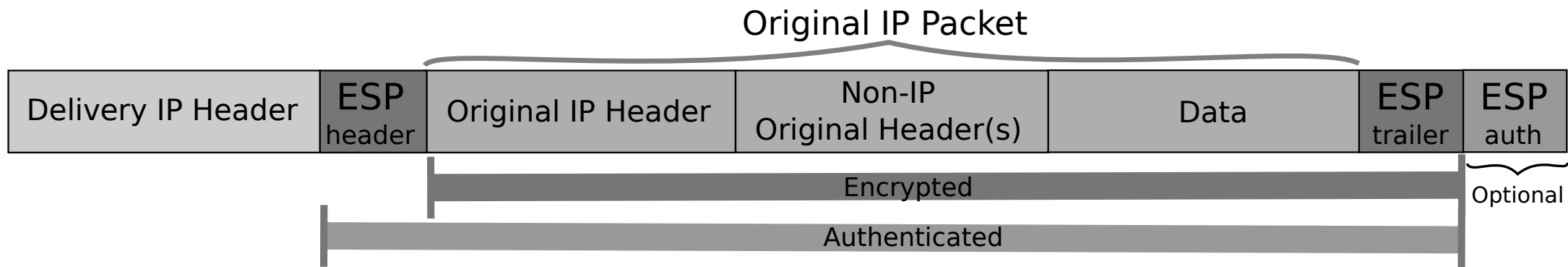


IPSec - ESP header placement

- Transport mode

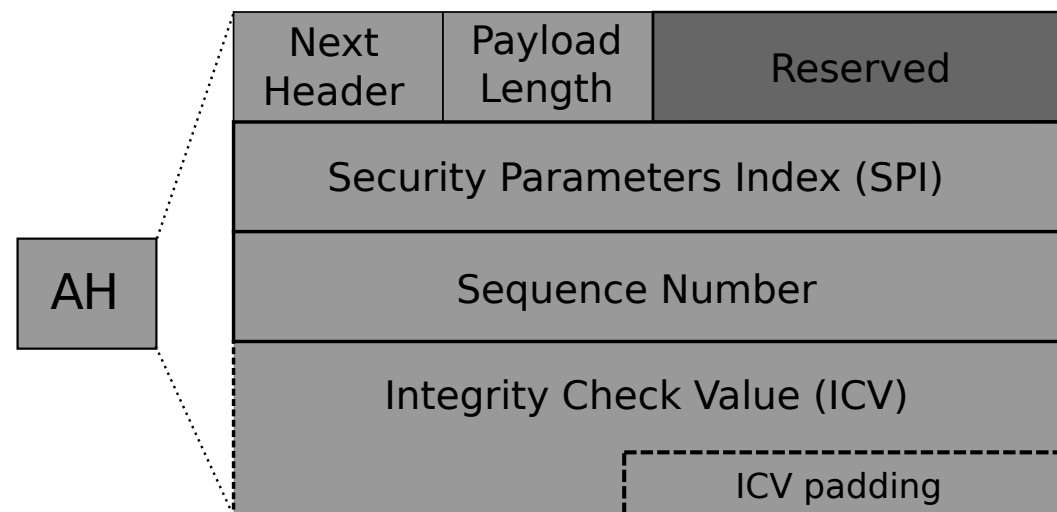


- Tunnel mode



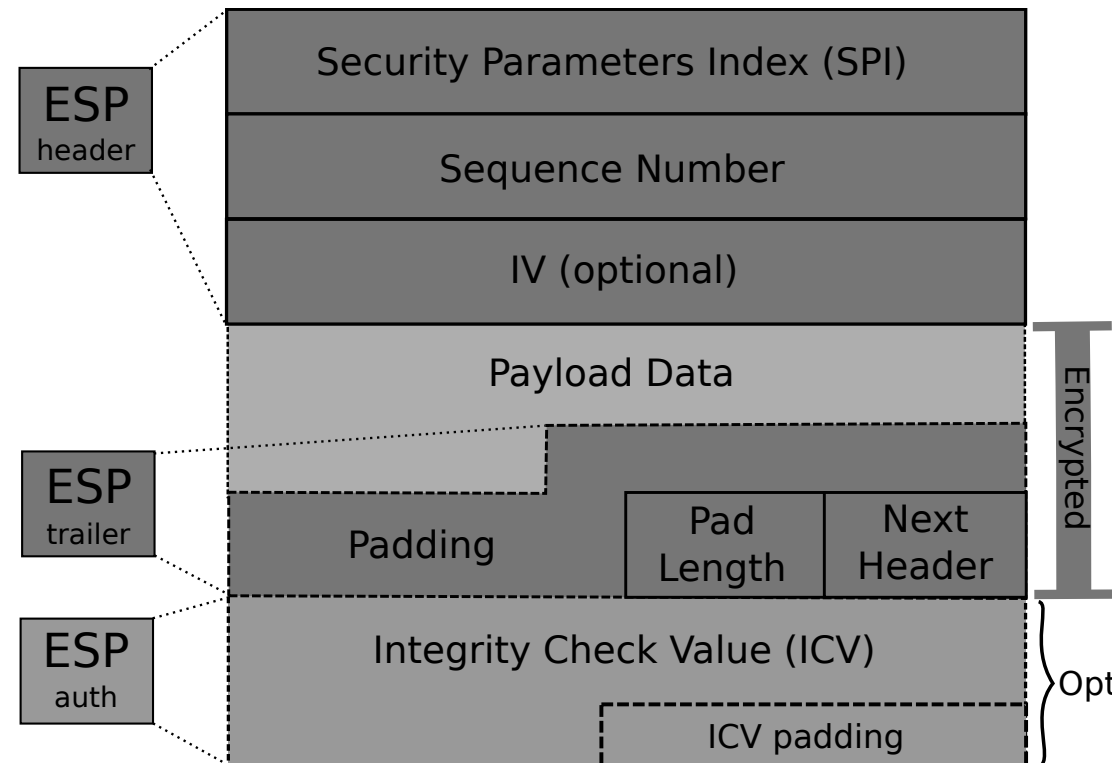
IPsec AH Header

- Contains five mandatory fields:
 - The Next Header field is an 8-bit field that identifies the type of the next payload after the AH.
 - The Payload Length is an 8-bit field specifying the length of the header (excluding the first 8 bytes) in 4-byte units.
 - The SPI field contains the negotiated outbound IPsec SPI and is used by the remote peer to identify the SA to which the packet belongs.
 - The Sequence Number field is a 32-bit field that contains a counter value that increases by one for each sent packet (using the same outbound IPsec SA).
 - The ICV field has a variable length (multiple of 32 bits) that contains the output of the authentication hash function (or HMAC based on symmetric encryption algorithms) applied to data/headers under protection.
 - ➔ May include padding to ensure that the overall length of the AH header is a multiple of 32 bits in IPv4 or 64 bits in IPv6.



IPsec ESP Header and Trailer

- Contain five mandatory fields:
 - The SPI field contains the negotiated outbound IPsec SPI and is used by the remote peer to identify the SA to which the packet belongs.
 - The Sequence Number field is a 32-bit field that contains a counter value that increases by one for each sent packet (using the same outbound IPsec SA).
 - The Padding field may contain 0 to 255 zero-bytes to guarantee: (i) a specific payload size imposed by the encryption algorithm (e.g., size multiple of the block cipher size), and (ii) that the Pad Length and Next header fields are right aligned within a 4-byte word.
 - The Pad Length is an 8-bit field that indicates the number of padding bytes in the Padding field.
 - The Next Header is an 8-bit field that identifies the type of data contained in the payload data.
- May contain two optional fields:
 - When the encryption algorithm requires an explicit Initialization Vector (IV), this value is sent using the IV field.
 - Some algorithm modes combine encryption and integrity into a single operation.
 - The ICV field has a variable length that contains the output of the authentication hash function (or HMAC based on symmetric encryption algorithms) applied to ESP header, Payload Data, and ESP trailer fields.
 - The ICV field may include padding.



IPSec - Security Associations

- SAs represent a policy contract between two peers or hosts
- Describe how the peers will use IPSec security services to protect network traffic
- An SA contains the following security parameters:
 - ♦ Authentication/encryption algorithm, key length and other encryption parameters (e.g. key lifetime, ...)
 - ♦ Session keys for authentication, or HMACs, and encryption, which can be entered manually or negotiated automatically
 - ♦ A specification of network traffic to which the SA will be applied (e.g. IP traffic or only TELNET sessions)
 - ♦ IPSec AH or ESP encapsulation protocol and tunnel or transport mode



Establishing SA and Cryptographic Keys

- ISAKMP - Internet Security Association and Key Management Protocol
 - ◆ Used to establishing Security Associations (SA) and cryptographic keys
 - ◆ Separate the details of security association management (and key management) from the details of key exchange
 - ◆ Provides a framework for authentication and key exchange but does not define them
- Oakley Key Determination Protocol
 - ◆ Key-agreement protocol
 - ◆ Allows authenticated peers to exchange keying material across an insecure connection
 - ◆ Uses Diffie-Hellman
- SKEME
 - ◆ Key exchange protocol
- IKE - Internet Key Exchange
 - ◆ Is a hybrid protocol
 - ◆ Uses part of Oakley and part of SKEME in conjunction with ISAKMP



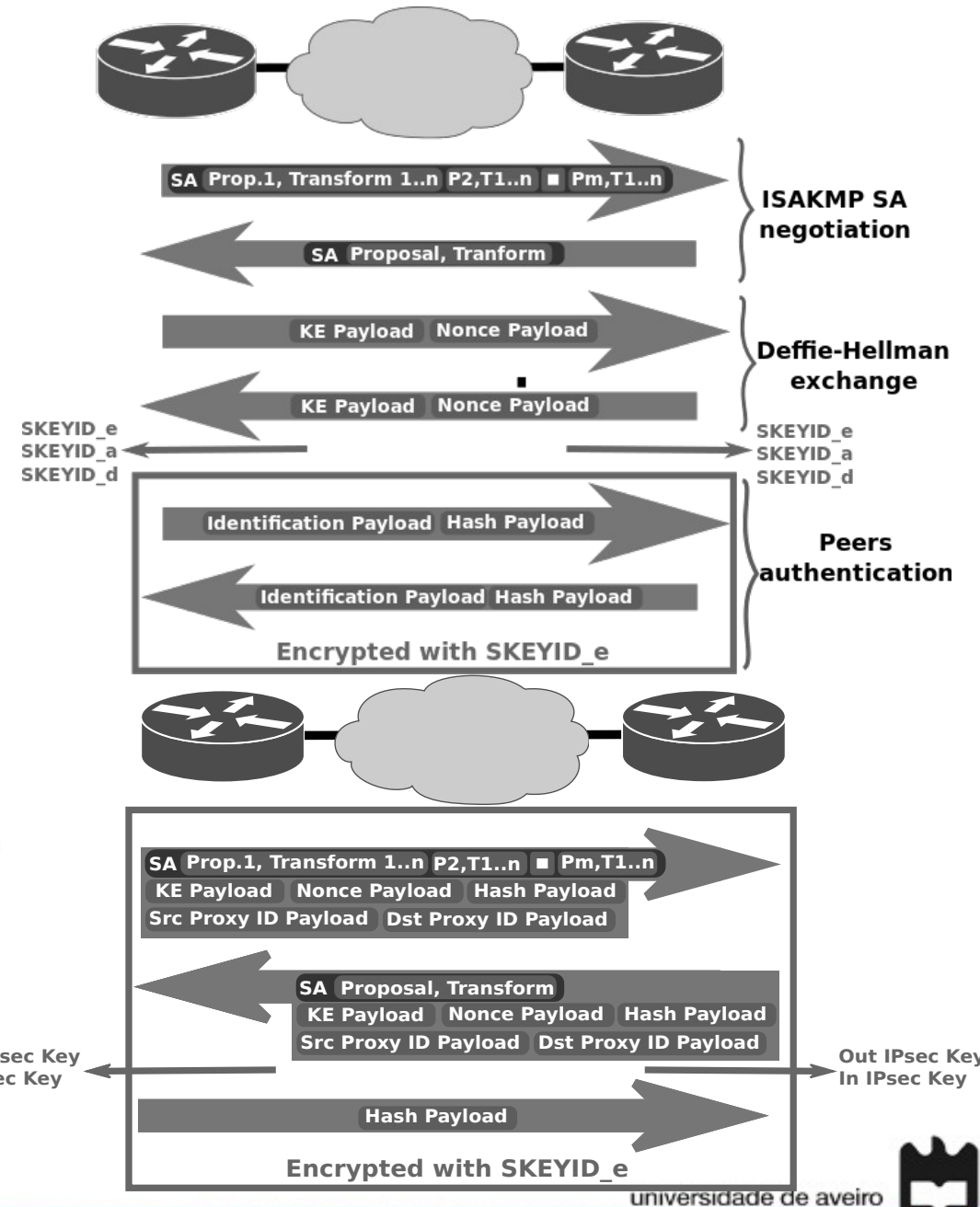
IKE/ISAKMP and IPsec

- Enhances IPsec by providing additional features and flexibility
- Provides authentication of the IPsec peers, negotiates IPsec keys, and negotiates IPsec security associations
- The IKE tunnel protects the SA negotiations. After the SAs are in place, IPsec protects data transference
- Advantages
 - ◆ Eliminates the need to manually specify IPsec security parameters at both peers
 - ◆ Allows administrators to specify a lifetime for the IPsec security association
 - ◆ Allows encryption keys to change during IPsec sessions
 - ◆ Allows IPsec to provide anti-replay services
 - ◆ Permits certification authority (CA) support for a manageable, scalable IPsec implementation
 - ◆ Allows dynamic authentication of peers
- IKE/ISAKMP provides three methods for two-way authentication:
 - ◆ Pre-shared key (PSK),
 - ◆ Digital signatures (RSA-SIG),
 - ◆ Public key encryption (RSA-ENC).



ISAKMP and IPsec – Phases/Modes

- ISAKMP modes control an efficiency versus security tradeoff during initial key exchange
- Phase 1
 - ◆ Peer agree on a set of parameters to be used to authenticate peers and to encrypt a portion of the phase 1 exchanges and all of phase 2 exchanges, authenticate peers, and generate keys to be used as generating keying material.
 - ◆ Main mode
 - ➔ Requires six packets back and forth
 - ➔ Provides complete security during the establishment of an IPsec connection
 - ➔ Aggressive mode is an alternative to main mode
 - Uses half the exchanges, but provides less security because some information is transmitted in cleartext
- Phase 2 - Quick mode
 - ◆ Peers negotiate and agree on parameters required to establish a fully functional IPsec communication service.



IPsec Packet Exchange

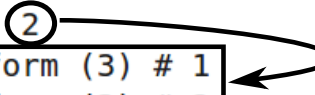
No.	Time	Source	Destination	Protocol	Length	Info
12	12.259744000	2001:a:a::2	2001:a:a::1	ISAKMP	146	Identity Protection (Main Mode)
13	12.293700000	2001:a:a::1	2001:a:a::2	ISAKMP	146	Identity Protection (Main Mode)
14	12.330320000	2001:a:a::2	2001:a:a::1	ISAKMP	298	Identity Protection (Main Mode)
15	12.364351000	2001:a:a::1	2001:a:a::2	ISAKMP	318	Identity Protection (Main Mode)
16	12.481540000	2001:a:a::2	2001:a:a::1	ISAKMP	170	Identity Protection (Main Mode)
17	12.496192000	2001:a:a::1	2001:a:a::2	ISAKMP	138	Identity Protection (Main Mode)
18	12.542122000	2001:a:a::2	2001:a:a::1	ISAKMP	250	Quick Mode
19	12.556571000	2001:a:a::1	2001:a:a::2	ISAKMP	250	Quick Mode
20	12.582568000	2001:a:a::2	2001:a:a::1	ISAKMP	114	Quick Mode
21	15.425134000	2001:a:a::2	2001:a:a::1	ESP	322	ESP (SPI=0xb26693bc)
22	15.440166000	2001:a:a::1	2001:a:a::2	ESP	202	ESP (SPI=0x328b3017)

▸ Frame 21: 322 bytes on wire (2576 bits), 322 bytes captured (2576 bits) on interface 0
 ▸ Ethernet II, Src: c2:04:62:06:00:00 (c2:04:62:06:00:00), Dst: ca:06:73:90:00:08 (ca:06:73:90:00:08)
 ▸ Internet Protocol Version 6, Src: 2001:a:a::2 (2001:a:a::2), Dst: 2001:a:a::1 (2001:a:a::1)
 ▾ Encapsulating Security Payload
 ESP SPI: 0xb26693bc (2993066940)
 ESP Sequence: 10



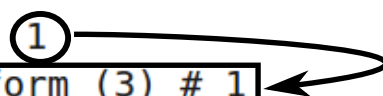
ISAKMP (phase 1) First Message

```
▷ Internet Protocol Version 4, Src: 200.2.2.2 (200.2.2.2), Dst: 200.1.1.1 (200.1.1.1)
▷ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▽ Internet Security Association and Key Management Protocol
  Initiator SPI: 06ba66b161c0b75d
  Responder SPI: 0000000000000000
  Next payload: Security Association (1)
▷ Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
▷ Flags: 0x00
  Message ID: 0x00000000
  Length: 204
▽ Type Payload: Security Association (1)
  Next payload: Vendor ID (13)
  Payload length: 96
  Domain of interpretation: IPSEC (1)
▷ Situation: 00000001
▽ Type Payload: Proposal (2) # 1
  Next payload: NONE / No Next Payload (0)
  Payload length: 84
  Proposal number: ①
  Protocol ID: ISAKMP (1)
  SPI Size: 0
  Proposal transforms: ②
    ▷ Type Payload: Transform (3) # 1
    ▷ Type Payload: Transform (3) # 2
▷ Type Payload: Vendor ID (13) : RFC 3947 Negotiation of NAT-Traversal in the IKE
▷ Type Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-07
▷ Type Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-03
▷ Type Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-02\n
```



ISAKMP (phase 1) Second Message

```
▸ Internet Protocol Version 4, Src: 200.1.1.1 (200.1.1.1), Dst: 200.2.2.2 (200.2.2.2)
▸ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▾ Internet Security Association and Key Management Protocol
  Initiator SPI: 06ba66b161c0b75d
  Responder SPI: 48aa62bdcbl9e9e3
  Next payload: Security Association (1)
▸ Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
▸ Flags: 0x00
  Message ID: 0x00000000
  Length: 104
▾ Type Payload: Security Association (1)
  Next payload: Vendor ID (13)
  Payload length: 56
  Domain of interpretation: IPSEC (1)
▸ Situation: 00000001
▾ Type Payload: Proposal (2) # 1
  Next payload: NONE / No Next Payload (0)
  Payload length: 44
  Proposal number: ①
  Protocol ID: ISAKMP (1)
  SPI Size: 0
  Proposal transforms: ①
  ▸ Type Payload: Transform (3) # 1
▸ Type Payload: Vendor ID (13) : RFC 3947 Negotiation of NAT-Traversal in the IKE
```



ISAKMP (phase 1) Third and Fourth Messages

▸ Internet Protocol Version 4, Src: 200.2.2.2 (200.2.2.2), Dst: 200.1.1.1
▸ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▼ Internet Security Association and Key Management Protocol

Initiator SPI: 06ba66b161c0b75d

Responder SPI: 48aa62bdcbl9e9e3

Next payload: Key Exchange (4)

▸ Version: 1.0

Exchange type: Identity Protection (Main Mode) (2)

▸ Flags: 0x00

Message ID: 0x00000000

Length: 276

▼ Type Payload: Key Exchange (4)

Next payload: Nonce (10)

Payload length: 132

Key Exchange Data: 6b90894c1593b8dddda8d321a05af8075

▼ Type Payload: Nonce (10)

Next payload: Vendor ID (13)

Payload length: 24

Nonce DATA: 21edc1d7ee9a9a51d9d8a0fccc1012ff9d58a348

▸ Type Payload: NAT-D (RFC 3947) (20)

▸ Type Payload: NAT-D (RFC 3947) (20)

▸ Internet Protocol Version 4, Src: 200.1.1.1 (200.1.1.1), Dst: 200.2.2.2
▸ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▼ Internet Security Association and Key Management Protocol

Initiator SPI: 06ba66b161c0b75d

Responder SPI: 48aa62bdcbl9e9e3

Next payload: Key Exchange (4)

▸ Version: 1.0

Exchange type: Identity Protection (Main Mode) (2)

▸ Flags: 0x00

Message ID: 0x00000000

Length: 296

▼ Type Payload: Key Exchange (4)

Next payload: Nonce (10)

Payload length: 132

Key Exchange Data: 820d0eafec6260bc958a60d1d086e6ec823032774f16c316...

▼ Type Payload: Nonce (10)

Next payload: Vendor ID (13)

Payload length: 24

Nonce DATA: 0f37423fb10f422983fcf0d9dcab26a5b8be59aa

▸ Type Payload: NAT-D (RFC 3947) (20)

▸ Type Payload: NAT-D (RFC 3947) (20)

ISAKMP (phase 1) Fifth and Sixth Messages

▷ Internet Protocol Version 4, Src: 200.2.2.2 (200.2.2.2), Dst: 200.1.1.1
▷ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▽ Internet Security Association and Key Management Protocol

Initiator SPI: 06ba66b161c0b75d

Responder SPI: 48aa62bdcdb19e9e3

Next payload: Identification (5)

▷ Version: 1.0

Exchange type: Identity Protection (Main Mode) (2)

▷ Flags: 0x01

Message ID: 0x00000000

Length: 92

Encrypted Data (64 bytes)

▷ Internet Protocol Version 4, Src: 200.1.1.1 (200.1.1.1), Dst: 200.2.2.2
▷ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▽ Internet Security Association and Key Management Protocol

Initiator SPI: 06ba66b161c0b75d

Responder SPI: 48aa62bdcdb19e9e3

Next payload: Identification (5)

▷ Version: 1.0

Exchange type: Identity Protection (Main Mode) (2)

▷ Flags: 0x01

Message ID: 0x00000000

Length: 68

Encrypted Data (40 bytes)



ISAKMP (phase 2) Message

```
▷ User Datagram Protocol, Src Port: 500 (500), Dst Port: 500 (500)
▽ Internet Security Association and Key Management Protocol
  Initiator SPI: 06ba66b161c0b75d
  Responder SPI: 48aa62bdc19e9e3
  Next payload: Hash (8)
▷ Version: 1.0
  Exchange type: Quick Mode (32)
▷ Flags: 0x01
  Message ID: 0x5277ae21
  Length: 220
  Encrypted Data
```

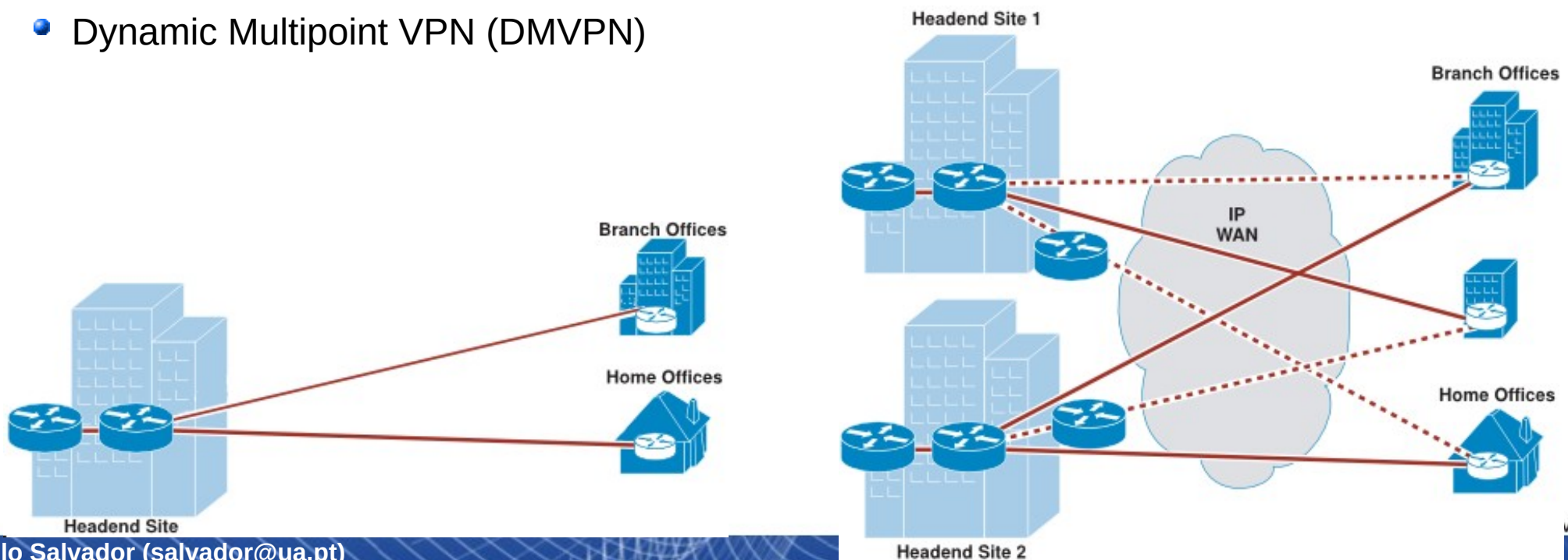


IPsec NAT Traversal

- NAT/PAT incompatibilities with IPsec
 - AH header incorporates the IP source and destination addresses in the keyed message integrity check. ESP is not an issue.
 - TCP and UDP checksums can be updated because are protected by IPsec.
 - IP addresses may be used as identifiers in Internet Key Exchange to determine credentials.
- During the ISAKMP IPsec first phase hosts (when configured and supported) detect that NAT traversal must be activated.
 - Subsequent ISAKMP first phase and second phase packets are encapsulated in UDP packets.
 - Usually port UDP 4500.
 - Original IP address are sent as NAT-OA (NAT Original Address) payloads of the ISAKMP.

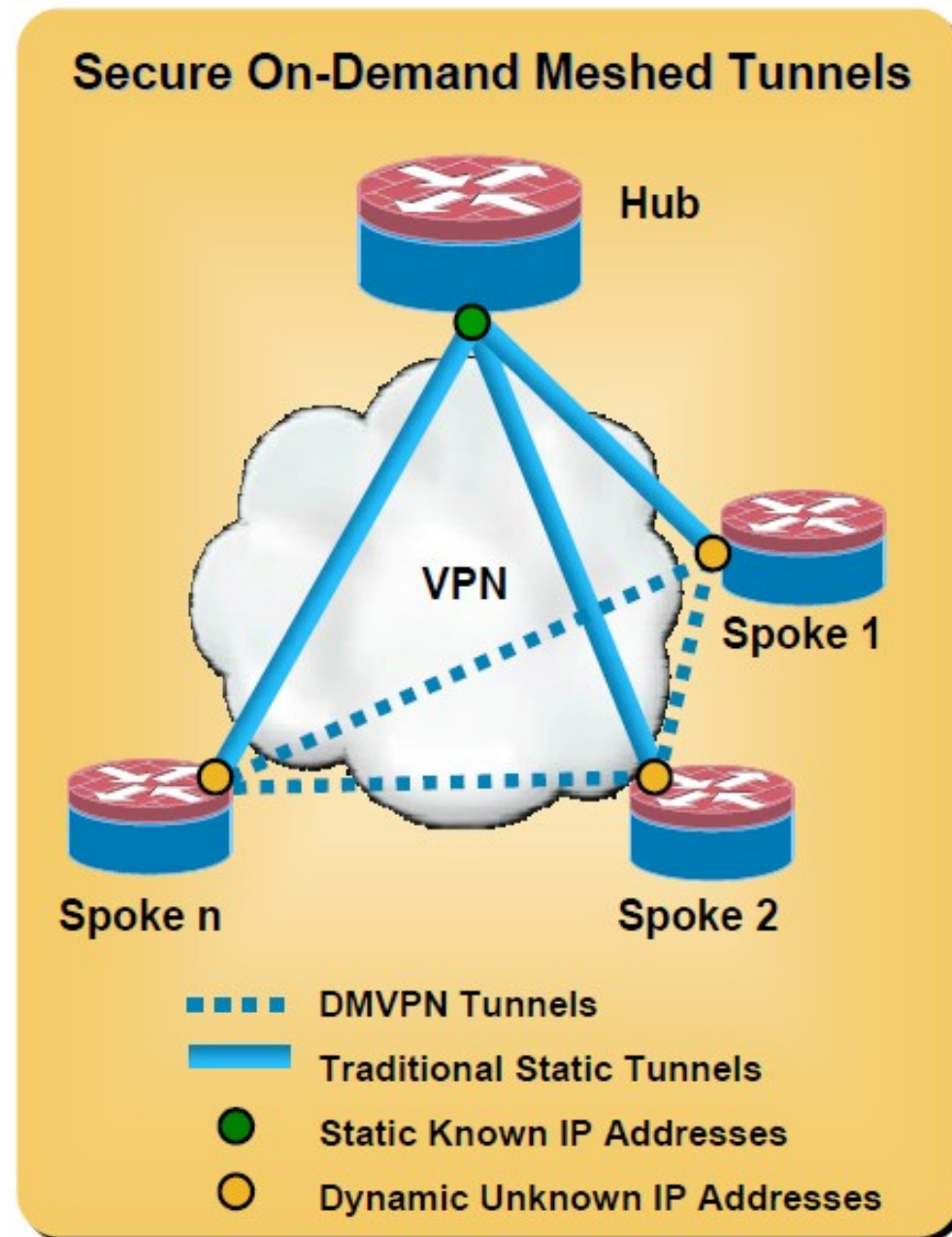
Variants of Site-to-Site IPsec VPN

- IPsec tunnels with static configuration
 - ◆ Requires the knowledge of all peers (IP addresses and security parameters)
 - ◆ High configuration overhead
- IPsec tunnels with dynamic configuration (at the headend/hub)
 - ◆ Hub + spokes configuration
 - ◆ Generic configuration at the headend/hub
 - ◆ Easy to add new spokes
- A basic IPsec tunnel can't protect multicast traffic.
- IPsec + GRE tunnels
 - ◆ Generic Routing Encapsulation (GRE) allows the protection of multicast traffic over IPsec
- Dynamic Multipoint VPN (DMVPN)



Dynamic Multipoint VPN

- Relies on NHRP to create overlay network
- Provides full meshed connectivity with simple configuration of hub and spoke
- Supports dynamically addressed spokes
- Facilitates zero-touch configuration for addition of new spokes
- Features automatic IPsec triggering for building an IPsec tunnel



SSL/TLS

- OpenVPN – Point-to-point
 - ♦ Uses either TCP or UDP and on any port (default 1194, or 443).
 - ♦ SSL/TLS stack (via OpenSSL) for key exchange.
 - ♦ Supports a multiple symmetric ciphers through the OpenSSL library.
 - Allows it to maintain backward compatibility with legacy hardware and use modern solutions.
- SoftEther
 - ♦ Multi-Protocol Engine
 - Can simultaneously listen for OpenVPN, IPsec, L2TP, WireGuard, etc...
 - ♦ Ethernet over HTTPS (Layer 2).
 - Usefull for remote access with Layer2 awareness (VPN number, VID).
 - ♦ Virtual Hubs and Bridging.
- Many other proprietary.



AnySec

- Still a Nokia-proprietary technology, working to be a standard.
- Uses IEEE802.1AE (MACsec) as its datapath encryption engine.
- and IEEE802.1x as its control plane signaling protocol.
- Provide an universal encrypted connection
 - ♦ In datacenter, providers and large enterprise networks.
 - ♦ Optimized for higher speeds (Terabits per second) and complex routing (MPLS, SR and SRv6).
- Alternative solution to:
 - ♦ MACsec, which is fast, but works on a point-to-point physical connection. Requires multi-hop independent deployment, which is extremely hard to manage.
 - ♦ IPsec, which works end-to-end, but is slow and adds overhead.



Cloud Based Solutions

- Also referred as SD-WAN or Cloud WAN.
- Also referred as VPN as a Service (VPNaaS).
- Network server/coordinator deployed in cloud.
- Data privacy concerns.
- Fully dependence on an external entity.
- Low initial investment.
- Operational costs difficult to anticipated beyond existing contracts.
- Multiple products from Cisco, Checkpoint, Fortinet, PaloAlto, Cloudflare, Tailscale (kind of cloud when using DERP nodes), etc...

